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A trustworthiness assessment for software based on multi-source evidence fusion

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ABSTRACT

Software trustworthiness assessment plays a critical role in ensuring the safety, reliability, and accountability of software systems, especially in safety-critical and mission-critical applications, which is inherently complex, as the multi-source evidence is used across various stages of the software life-cycle. To address the resulting heterogeneity of evidence sources, this paper proposes a multi-stage and multi-source trustworthy evidence fusion method. First, a conflict measurement method between direct and indirect evidence is developed based on the concept of falsity. Second, a conflict resolution strategy for homogeneous evidence is proposed by integrating credibility and uncertainty metrics. Finally, a weighted combination method grounded in game theory is proposed to integrate falsity, credibility, and uncertainty, followed by evidence fusion using the Dempster rule. Experimental results in real software systems assessment demonstrate that the proposed model and algorithm not only satisfy the fundamental principles of trustworthiness assessment, namely monotonicity, acceleration, sensitivity, and substitutability, but also achieve higher accuracy and robustness compared with state-of-the-art algorithms.

1. Introduction

Software trustworthiness refers to the capability of a software system to fulfill user expectations through its behaviors and outcomes, while maintaining continuous service in the presence of disturbances or failures [1]. Trustworthiness assessment frameworks typically integrate multiple essential software characteristics-covering the trustworthiness attributes proposed in [2–7]-to construct a multidimensional assessment model. These models quantify the actual performance of each attribute and apply appropriate weight configurations to build a computable framework, which provides a rigorous basis for assessing the system's operational stability, service reliability, and security assurance. Such a quantitative assessment mechanism serves not only as a core component of software quality assurance, but also plays a crucial role in system risk prediction and performance assessment. As a result, the quantification of software trustworthiness has become a central scientific issue in the field of system reliability research.

In the field of software trustworthiness assessment, various models have been proposed to address different aspects of the assessment process. Tao et al. [8] first introduced an attribute-based trustworthiness measurement model, which laid the foundation for subsequent research. Building on this work, Tao and Chen [9] developed a dynamic trustworthiness assessment model that adjusts the assessment results in real time according to changes in software runtime attributes. Tian and Guo [10] considered specific operational behaviors of software and proposed a behavior trajectory matrix-based assessment model, which assesses trustworthiness by analyzing execution patterns during system operation.

The aforementioned models typically adopt a bottom-up structure that decomposes software trustworthiness into attributes, sub-attributes, and supporting evidence. However, they often treat evidence as a single homogeneous category, overlooking the differences in evidence sources and their contextual characteristics. In practice, software trustworthiness assessment is not merely a process of quantifying individual attributes but rather a comprehensive and multi-layered procedure. Each phase of the software development life-cycle produces distinct types of evidence, originating from various activities and stakeholders. As illustrated in Fig. 1, evidence is generated throughout stages such as requirements analysis, system design, implementation, testing, deployment, maintenance, and user feedback. These phases involve diverse decision-making and development tasks, each of which

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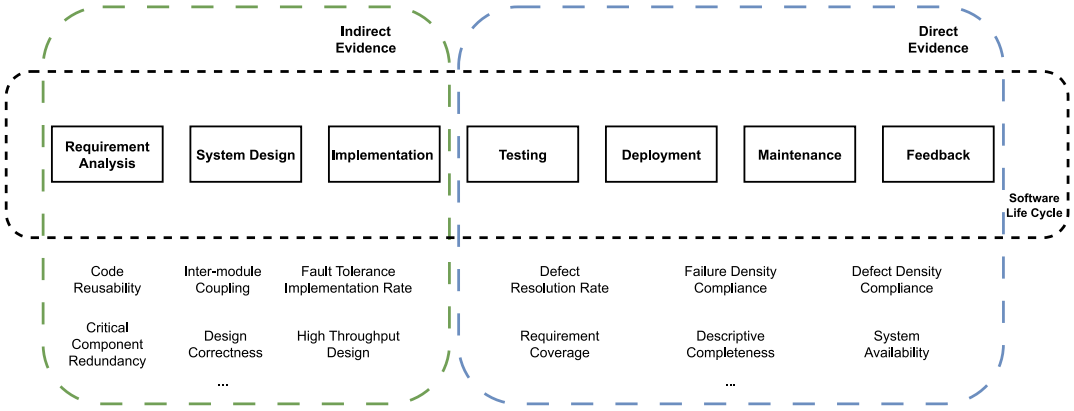


Fig. 1. Software Life-cycle and Sources of Evidence.

may impact the final trustworthiness of the software to varying degrees. For instance, during the requirements analysis phase, evidence such as code reusability, inter-module coupling, and redundancy of critical components may be collected; whereas in the testing phase, data including defect resolution rates and fault density compliance are typically gathered as key inputs for trustworthiness assessment. Such evidence includes not only design and implementation documents provided by developers, but also operational data obtained from testers, maintainers, and end users.

Moreover, the types of evidence involved are highly diverse, including both quantitative metrics (e.g., fault density, defect resolution rate) and qualitative descriptions (e.g., design correctness, system availability). As software development progresses, the type, source, and contextual influence of the evidence continuously evolve. For example, evidence obtained in the early stages based on design documents may differ significantly from failure rate data collected during the testing phase. Such discrepancies may lead to conflicts in the assessment process, especially when fusing evidence across different stages. Variations in evidence source, type, and quality often introduce inconsistencies and conflicts, further increasing the complexity of trustworthiness assessment.

Therefore, a key challenge in software trustworthiness assessment lies in how to effectively integrate evidence collected from multiple development stages and various stakeholders, while appropriately resolving the conflicts that may arise among them. As the assessment process deepens, the ability to distinguish and handle different types of evidence, particularly indirect evidence originating from various stages of the development process and direct evidence derived from runtime data and user feedback, has become a central focus of this paper. Specifically, the limitations of existing trustworthiness assessment models are mainly reflected in two aspects:

First, current models fail to adequately consider the source, type, and fusion method of evidence associated with trustworthiness attributes. For example, Ref. [11] focuses solely on reallocating software component development costs to optimize trustworthiness, while neglecting the diversity of evidence sources involved in trustworthiness assessment, resulting in a partial and limited evidence collection and fusion strategy. Ref. [12] considers component construction methods, weight assignment, and user feedback to assess system trustworthiness, but overlooks how to effectively integrate multi-source heterogeneous evidence, particularly the differential contributions of various evidence types to the final assessment. Ref. [13] provides trustworthiness support during source code architecture and verification but does not address fusion methods across different types of evidence. As shown in Fig. 1, evidence related to trustworthiness attributes originates from multiple stages of the software life-cycle. Runtime testing results and user feedback can directly reflect the current state of the software's trustworthiness. In contrast, evidence collected during the development process, such as code quality or module reuse rate, may indicate the long-term evolution trends and potential risks of the software. For instance, a software system may perform well during testing but experience frequent requirement changes during development. Existing models may assign a high trustworthiness score based on short-term outcomes, while overlooking potential risks that may emerge during future maintenance phases.

Second, existing models lack effective mechanisms for handling conflicts and complementarity in the fusion of multi-source and multi-type evidence. For instance, Ref. [14] only addresses how to assign weights to software component attributes, with the model primarily focusing on the quantification and weighting of single-layer attributes, without considering fusion strategies among evidence from different sources. Ref. [15] proposes a trustworthiness allocation and reallocation method within a mathematical programming framework, but does not delve into how to effectively resolve conflicts or adjust trustworthiness when inconsistencies or complementarities exist among evidence items. Conflicts may arise when evidence originates from different sources. For example, evidence from the testing phase may indicate that all functions pass validation, while development-phase evidence may reveal that code complexity exceeds acceptable thresholds. Most existing models apply simple weighted averaging or linear aggregation methods, which are incapable of identifying and correcting such conflicts, thereby undermining the reliability of the assessment results. Additionally, evidence from different sources may be complementary. In safety-critical systems, for example, evidence collected during testing may fail to expose latent vulnerabilities due to insufficient test coverage, whereas development-phase evidence may provide supplementary insights. However, such complementarity is often overlooked by existing models, resulting in the underutilization of valuable information.

To address the above limitations, the trustworthiness assessment model proposed in this paper classifies evidence into two categories: direct evidence and indirect evidence. Direct evidence is derived from system testing, user trials, and feedback, and is considered authoritative as it directly reflects the current trustworthiness level of the software. Indirect evidence, on the other hand, originates from earlier development phases such as system design and code implementation. Examples include module reuse rate and code complexity, which typically indicate the quality control and management practices during development. Such indirect evidence is used to assess the process-level trustworthiness of the software.

Based on the source and type of evidence, this paper proposes a multi-stage trustworthy evidence fusion method, which includes the following components: (1) Conflict correction between different types of evidence: A falsity-based method is proposed to quantify the conflict between indirect and direct evidence by calculating the falsity degree of

each, thereby identifying and mitigating their inconsistencies. (2) Conflict correction within the same type of evidence: For evidence within the same category (either indirect or direct), credibility and uncertainty metrics are introduced to resolve internal conflicts among homogeneous evidence sources. (3) Fusion of multiple types of evidence: A linear combination method based on game theory [16] is proposed to transform conflicts, both between direct and indirect evidence and within the same type, into a weighted adjustment model. The adjusted evidence values are then fused using the Dempster rule, ensuring that the final trustworthiness assessment is both accurate and robust.

Compared with existing methods, the main contributions of this paper are summarized as follows:

(1) A software trustworthiness assessment model is proposed, which explicitly distinguishes between direct and indirect evidence based on their sources, addressing the limitation of conventional models that treat evidence as a homogeneous category. The model not only emphasizes the outcome trustworthiness reflected by runtime data but also explores the process trustworthiness embedded in development-phase artifacts.

(2) A multi-stage trustworthy evidence fusion method is proposed, which resolves conflicts caused by inconsistencies among evidence from different sources. The method adopts a falsity-based strategy for heterogeneous evidence and a credibility-uncertainty-based strategy for homogeneous evidence. A linear combination strategy based on game theory is used for dynamic weight allocation and optimization of composite discounting factors, ensuring the mathematical rigor as well as the rationality and consistency of evidence adjustment.

(3) Three core research questions are designed in experiment. Through case studies and comparative experiments, the proposed model is shown to fulfill basic assessment criteria such as monotonicity, acceleration, sensitivity, and substitutability, while delivering improved performance in terms of both accuracy and robustness.

This paper is organized as follows. Section 2 reviews related work on trustworthiness assessment methods and evidence conflict handling approaches. Section 3 presents the theoretical foundations of evidence conflict resolution in the context of software trustworthiness assessment. Section 4 proposes a software trustworthiness assessment model based on the classification framework of direct and indirect evidence, and proposes a multi-stage evidence fusion method. Section 5 describes the experimental design and reports the results. Section 6 concludes the paper and discusses directions for future work.

2. Related work

2.1. Software trustworthiness models

In the field of software trustworthiness evaluation, extensive research has been conducted, particularly focusing on modeling approaches and assessment methodologies. Among the mainstream approaches, attribute-based and component-based modeling methods are widely studied and applied.

Attribute-based modeling views software trustworthiness as the result of interactions among multiple interrelated attributes. These attributes influence the overall trustworthiness evaluation in a nonlinear and synergistic manner, especially in safety-critical domains such as aerospace. Tao et al. [17] proposed a trustworthiness evaluation model based on key attributes, demonstrating that it should satisfy four fundamental criteria: monotonicity, acceleration, sensitivity, and substitutability. Building upon this, Ref. [18] introduced an axiomatic method that classifies attributes into critical and non-critical categories for quantitative assessment. This model showed strong performance in evaluating 23 aerospace software systems, effectively identifying weak points in the development process. Further, Ref. [19] extended the attribute set and enriched the theoretical foundation by introducing new properties such as non-negativity, proportionality, full substitutability, and expectability. To improve practicality and accuracy, Ref. [20] proposed an enhanced partition-based method that optimized the original model.

Component-based modeling, on the other hand, assesses system trustworthiness by differentiating between critical and non-critical components. Wang et al. [21] suggested dividing software components based on their impact on core functionality, where failures in critical components could lead to system-wide failures, while failures in non-critical components had limited impact. This classification facilitates component reuse, reduces development time and cost, and simplifies system management [22,23]. A practical example is provided in Ref. [24], where in a second-level parallel structure of an aerospace prototype system, failure of a propeller unit (critical) could halt system operation, while failure of a splashdown unit (non-critical) would not result in catastrophic outcomes.

However, although existing assessment models have provided detailed analyses of software trustworthiness attributes and component importance, they often overlook the issue of evidence classification across the full software life-cycle. In particular, evidence originating from different development stages plays significantly different roles in trustworthiness assessment. For example, evidence produced during requirements analysis and system design tends to be more theoretical, whereas evidence obtained during testing and maintenance is often more practical and dynamic. Most existing models treat all evidence as a single category, ignoring the complementarity and conflicts that may exist among different types of evidence. In reality, the source and type of evidence directly influence the final trustworthiness score, especially in complex assessment scenarios involving multiple stages and heterogeneous sources. Therefore, trustworthiness assessment models should place greater emphasis on lifecycle-aware evidence classification and design appropriate evidence fusion strategies. Such improvements are essential to effectively manage conflicts and leverage complementarities among diverse evidence, thereby enhancing both the accuracy and robustness of assessment results.

2.2. Evidence fusion algorithms

In the field of software trustworthiness assessment, several studies have proposed evidence-based fusion frameworks aimed at improving the rationality and automation of the assessment process. Deng et al. [25] introduced the Trust Derivation Tree model, which employs first-order logic rules to iteratively decompose high-level trustworthiness attributes into low-level evidence, and utilizes Prolog for logical reasoning to support structured modeling and verification of software trustworthiness. Li [26] proposed a collaborative trustworthiness assessment model based on real-world evidence, which quantifies data collected across the software lifecycle to construct a trust level model, thereby improving the model's applicability in practical scenarios. Wang et al. [27] developed a rule-driven trustworthiness assessment framework, which constructs trust trees based on configurable rules that encapsulate data collection protocols, format definitions, and measurement algorithms. The model further enhances interpretability and diagnostic capabilities through level-matching and bottleneck analysis algorithms.

Further, Bao [28] developed a Trustworthiness Degree Model based on a dual-set constraint of necessary and optional evidence. The model employs a layered threshold mechanism and an AHP-based weighting strategy to integrate multi-attribute evidence, aiming to enhance the granularity and reliability of assessment and feedback. Li and Chen et al. [29] proposed a trustworthiness measurement model based on weighted information entropy, which emphasizes the use of entropy to quantify uncertainty and assess the program anomalies indicated by trustworthy evidence. In addition, Ding [30] introduced the Objective Weight based Evidential Reasoning method, which integrates objective weight computation with the Evidential Reasoning framework. By incorporating indicators such as total uncertainty and confidence level,

the method addresses the limitations of traditional Software Trustworthiness assessment models that rely heavily on subjective judgment and suffer from high uncertainty.

Although the above methods have made significant progress in fusion strategies, reasoning models, and weight assignment, a key limitation remains in their handling of conflicts among multi-source heterogeneous evidence. Most approaches treat all evidence as uniform input without systematic classification. They lack a clear distinction between evidence types and sources, and do not establish a stage-aware fusion mechanism to address conflicts or complementarities between different categories of evidence. This omission becomes particularly critical in complex systems, where the fusion process may become overly sensitive to high-conflict or low-quality evidence, ultimately compromising the reliability of trustworthiness assessments.

On the other hand, since the introduction of Dempster–Shafer evidence theory, the framework has demonstrated significant application value across various domains due to its unique strength in handling uncertain information [31]. In structural reliability, Zhang et al. [32] introduced a spline-based probabilistic transformation to improve assessment under uncertainty, while Toppila and Salo [33] proposed an interval-based modeling approach for robust system enhancement decisions, and more recent studies further advanced dynamic reliability assessment through evidential reasoning and convex-model-based approaches [34,35]. For health management and prognostics, Xu et al. [36] developed a multi-source evidence reasoning model with entropy-based weighting and genetic optimization, and Certa et al. [37] used evidence theory to integrate expert inputs for more robust failure ranking; subsequent works have also extended evidential reasoning rules with dynamic correlation and hybrid causal logic to improve inference accuracy in complex risk contexts [38,39]. In intelligent systems, Liu and Zhang [40] applied basic probability assignment and Dempster's rule to improve decision reliability in voting mechanisms. Helton and Johnson [41] demonstrated evidence theory's potential in nuclear safety modeling, and Shah et al. [42] extended it to mixed uncertainty settings via polynomial chaos expansions. Within railway systems, Liu et al. [43] modeled fault propagation using component impactability and susceptibility metrics, and Li et al. [44] evaluated system resilience under disruptions through hybrid fuzzy-evidence-Bayesian reasoning, with more recent work applying evidential reasoning to assess resilience in ports and cross-border freight railway systems [45–47]. Additionally, Zhang et al. [48] constructed an evidential network to infer system reliability from linguistic data, while Eldred et al. [49] built a hybrid uncertainty quantification framework suitable for large-scale engineering applications.

Although D-S theory has achieved notable success in the aforementioned domains, its application in software trustworthiness assessment remains limited. Existing studies primarily focus on attribute weight assignment, without fully exploring the potential of D-S theory in handling multi-source heterogeneous evidence fusion, conflict resolution, and life-cycle-wide evidence fusion.

To address these limitations, this paper proposes a multi-stage trustworthy evidence fusion method based on evidence source and type. By explicitly distinguishing between indirect and direct evidence, the method constructs a staged conflict identification and correction mechanism. It further integrates three key indicators: falsity, credibility, and uncertainty, which are used for weighted fusion. This method enables more accurate handling of evidence from different stages and sources, thereby enhancing the assessment capability and robustness of the model in multi-source environments.

3. Preliminary

This section begins with utility theory, which is employed to preprocess trust-related evidence by transforming it into a form of basic belief assignment (BBA) within a defined frame of discernment. Then, the section introduces the Dempster–Shafer theory of evidence, which provides the mathematical framework for representing and combining uncertain and imprecise information. Next, conflict computation methods are discussed to quantify the degree of disagreement between different evidence sources, laying the groundwork for conflict management strategies. Finally, the concept of negation of BBA is introduced to enhance the model's flexibility in representing conflicting information. These theoretical components collectively provide the basis for the multi-stage evidence fusion method proposed in Section 4.

3.1. Utility theory

During the process of software trustworthiness assessment, a wide variety of evidence is collected and measured according to different assessment criteria. To ensure consistency in scale and facilitate subsequent data fusion, a utility-based data normalization method is commonly adopted to standardize the measured values of trustworthiness evidence [50].

For qualitative indicators, the utility value is computed using a weighted utility function that incorporates expert-defined confidence levels. For quantitative indicators, a piecewise utility function is employed to map raw values to corresponding utility scores. Based on these mappings, the utility value of each indicator is determined, and its confidence levels with respect to assessment utility grades are derived through interpolation.

Furthermore, once the mass functions have been calculated through evidence fusion, the resulting utility values can be converted back into interpretable measurement outputs to support decision-making. The detailed mathematical definitions and transformation formulas used in this process are provided in Appendix A.1.

3.2. Dempster-Shafer theory

Dempster–Shafer theory, originally developed by Dempster and further refined by Shafer [51,52], provides a theoretical framework for handling uncertainty. Compared to classical probability theory, it relaxes some of the strict axiomatic requirements. When precise probabilistic information is available, evidential reasoning converges to probability theory. However, in situations where prior probabilities are difficult to obtain, it demonstrates distinct advantages and applicability.

Definition 1. Let θ denote the domain set of all possible values of F, where all elements within θ are mutually exclusive. Then, θ is referred to as the frame of discernment for F.

Definition 2. Let θ be a frame of discernment. A function $m : 2^{\theta} \rightarrow [0, 1]$ is called a basic belief assignment (BBA) if it satisfies the following conditions:

(1) $m(\emptyset) = 0$,

(2) $\sum_{A \in F} m(A) = 1$

Then $m(A)$ is called the BBA of A, representing the degree of belief assigned to A. If $m(A) > 0$, A is referred to as a focal element. In evidence theory, $Bel(A)$ is known as the belief function (BF) or lower bound function, indicating the degree of belief that A is true. $Pl(A)$ is called the plausibility function (PF) or upper bound function, representing the degree to which A is not false.

Definition 3. Let Bel_1 and Bel_2 be two belief functions defined on the same frame of discernment θ, with corresponding basic belief assignments m_1 and m_2, and focal elements $P_1, P_2, \ldots, P_i$ and $Q_1, Q_2, \ldots, Q_j$, respectively.

$$m_{1,2}(Z) = \begin{cases} 0, & Z = \emptyset \\ \dfrac{\sum_{i,j,\ P_i \cap Q_j = Z} m_1(P_i) m_2(Q_j)}{1 - K}, & \forall Z \subset U,\ Z \neq \emptyset \end{cases} \tag{1}$$

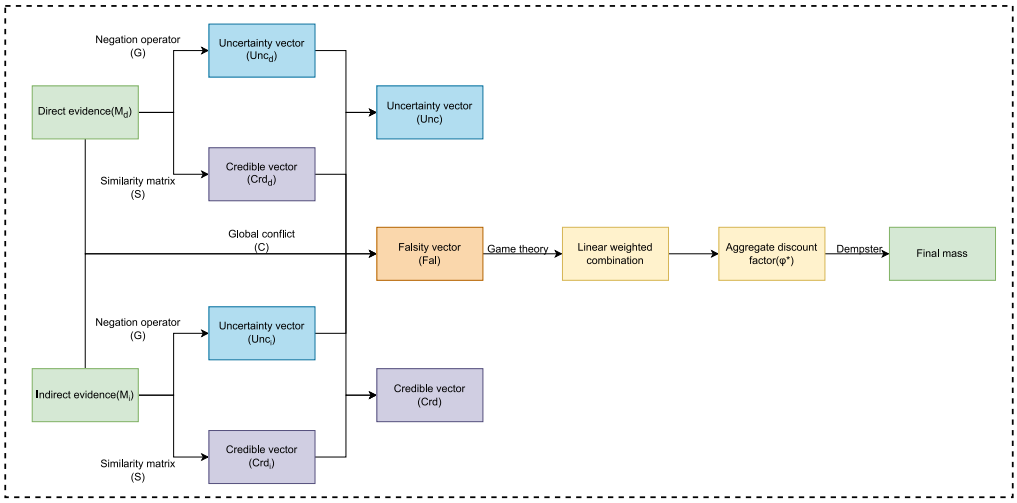


Fig. 2. A Multi-stage Trustworthy Evidence Fusion Method.

where $K = \sum_{i,j,\ P_i \cap Q_j = Z} m_1(P_i)\, m_2(Q_j) < 1$ is the conflict coefficient. If $K \neq 1$, the combined mass function mm defines a valid basic belief assignment. If $K = 1$, it indicates that m_1 and m_2 are in complete conflict and therefore cannot be combined.

The classical Dempster–Shafer (DS) evidence theory performs well when fusing evidence with low conflict. However, when the evidence is highly conflicting or directly contradictory, its fusion capability deteriorates, often producing counterintuitive results. Typical issues include the total conflict paradox, zero-belief paradox, full-belief paradox, evidence invalidation paradox, and belief deviation paradox.

3.3. Evidence conflict measurement methods

To facilitate conflict quantification in multi-source evidence fusion, this section introduces three representative measures based on Dempster–Shafer theory. Shafer's conflict measure [53] calculates the total belief mass assigned to mutually exclusive focal elements during conjunctive combination. While effective for basic conflict analysis, it only considers intersections that result in the empty set, thus overlooking partial conflicts. To address this, the Jousselme distance [54] incorporates the similarity between focal elements to quantify the dissimilarity between two BBAs more comprehensively. Building upon these, the binary Shafer's conflict measure [55] combines classical conflict mass with the maximum pairwise Jousselme distance and Jaccard similarity, enabling a more complete representation of both direct and indirect conflicts. The mathematical definitions of these measures are provided in Appendix A.2.

3.4. Negation of BBA

To quantify the uncertainty between pieces of evidence, the negation of BBA is introduced. In the Dempster–Shafer theory of evidence, the negation of BBA is an operation that reconfigures the belief structure to reflect epistemic uncertainty about the non-validity of propositions [56]. This transformation tends to increase the entropy of the BBA, resulting in a more uniform belief distribution and reflecting a reduced level of confidence in the system's current state.

The negation of a BBA satisfies several important properties:

(1) Every BBA has a unique negation BBA.

(2) The negation assigns a value to the empty set, denoted as $m_{not}(\emptyset)$, which reflects the exhaustiveness of the frame of discernment θ.

(3) If $m(A) = 1$, $m_{not}(A) = 0$, where $A \subseteq \theta$.

(4) If the entropy of a BBA is defined as H, then $H(m) \leq H(m_{not})$. According to Ref. [57], the negation operation is irreversible and therefore leads to an increase in information entropy.

For computational purposes, the BBA can be represented as a column vector (BBAV) over the power set of the frame of discernment. The negation operation is then implemented via a matrix transformation, where a specially constructed negation operator is applied to the BBA vector. This operator redistributes belief mass according to the set-theoretic relationships between focal elements and their complements. The mathematical definitions and matrix construction rules for the negation operator are provided in Appendix A.3.

4. A multi-stage trustworthy evidence fusion method

Conflicts among evidence are inevitable in the process of software trustworthiness assessment. These conflicts mainly stem from the diversity of data sources and the differences in acquisition methods. Software assessment typically relies on various sources, such as development process documentation, standardized test results from laboratories, and runtime data collected from operational environments. Variations in acquisition settings, measurement tools, and assessment criteria across these sources often result in inconsistent representations of the same attribute. Moreover, objective differences between testing environments and real-world deployment scenarios, such as hardware configurations, network conditions, and system load, can lead to discrepancies in indicator values, further contributing to evidence conflict. In addition, data uncertainty and quality issues play a significant role. During data collection, factors such as measurement errors, noise, missing values, and subjective judgment can degrade the accuracy and consistency of the evidence.

To address above issues, a scientifically sound conflict resolution strategy is essential during evidence fusion. This paper proposes a multi-stage trustworthy evidence fusion method, as illustrated in Fig. 2. First, based on the classification framework presented in Section 4, all evidence is categorized by source into direct and indirect evidence. Then, the global conflict between these two categories is quantified by calculating their respective falsity degrees. Next, the internal conflict within each category, whether among direct or indirect evidence, is assessed using credibility and uncertainty measures to capture local inconsistencies. Finally, a game-theoretic linear combination method is applied to integrate the various levels of conflict indicators into a unified weighted model. This yields a composite discounting factor that is used to adjust individual evidence values, ensuring that the final fusion result is both more accurate and more robust.

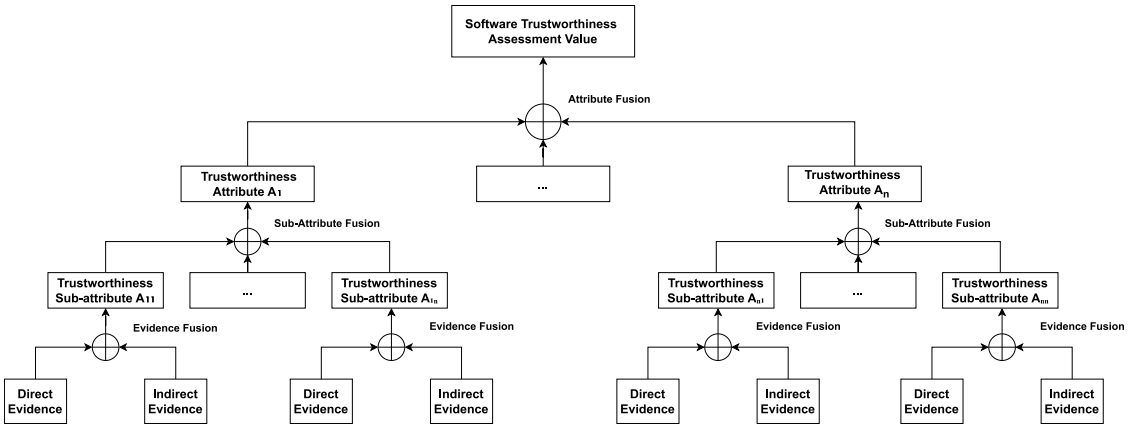


Fig. 3. The Proposed Software Trustworthiness Model.

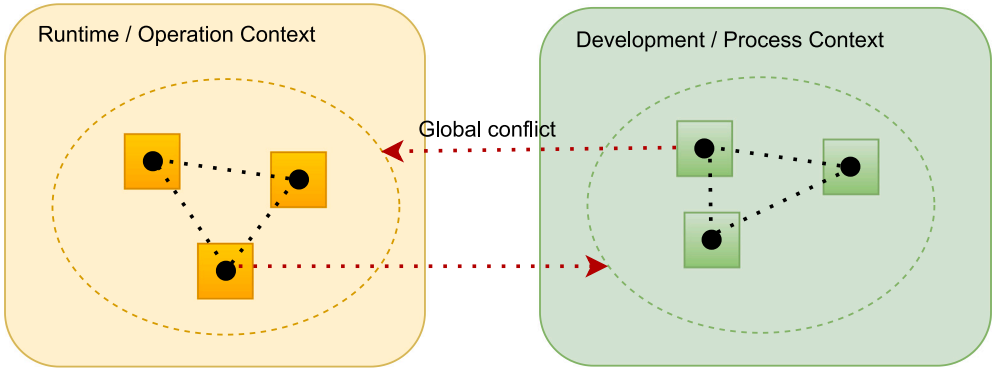


Fig. 4. Semantic heterogeneity of evidence: Direct vs. Indirect.

4.1. A trustworthiness model based on direct and indirect evidence classification

To address the inconsistency of evidence sources in software trustworthiness assessment, this paper builds upon attribute-based assessment models and proposes a novel model that classifies evidence into direct and indirect types. As illustrated in Fig. 3, the model adopts a hierarchical fusion architecture, progressively integrating information from the evidence layer, through the sub-attribute layer, up to the attribute layer, to ultimately derive the final trustworthiness assessment result. The trust-related evidence primarily originates from well-established trust metrics widely adopted in both academia and industry.

Definition 4. *Evidence=(EName, EType, Performance)*, where *EName* denotes the name of the evidence, *EType* refers to the type of evidence, and *Performance* represents its measured value. Given the wide variety of available trustworthiness evidence in software systems and the fact that many are mutually substitutable (e.g., defect rate vs. defect density), the model does not mandate the use of specific indicators. Instead, it defines general requirements for evidence selection and highlights common choices. The examples used in this paper are primarily drawn from ISO/IEC 25010 and widely adopted industrial metrics.

As shown in Table 1, software trustworthiness evidence can be divided into two semantically distinct types:

Direct evidence is typically obtained through actual software operation and runtime outcomes. It reflects the result-oriented state and performance of the software and is used to assess outcome-level trustworthiness. For example, by conducting functional tests, assessors can directly determine whether the software functions meet the specifications outlined in the requirements documentation, thereby evaluating the software's current reliability and effectiveness.

Indirect evidence originates from the software development process and is generally extracted from artifacts such as design specifications, requirements documents, or process-related metrics (e.g., code complexity, reuse rate). This evidence does not measure operational outcomes directly, but rather characterizes process-level properties that serve as proxies for predicting potential reliability.

As shown in Fig. 4, owing to this semantic heterogeneity, conflicts between direct and indirect evidence differ in nature from those among homogeneous evidence sources. According to the interpretation in Ref. [58], such cross-type conflicts are more appropriately regarded as representation errors, indicating that at least one piece of evidence does not belong to the same frame of discernment, hence falsity is employed to quantify global inconsistencies, while credibility and uncertainty are reserved for resolving intra-type local inconsistencies.

This paper proposes a trustworthiness assessment model based on the classification of direct and circumstantial evidence. The model aims to comprehensively reflect the influence of evidence sources and their quality characteristics on reliability-related sub-attributes. Specifically, we categorize the data into direct and circumstantial evidence according to their sources, and assess them separately based on their inherent quality features. After establishing the overall trustworthiness assessment framework, we introduce the underlying theory for handling evidence conflicts, which serves as the foundation for the subsequent analysis. The detailed implementation method of the proposed assessment model, including the conflict handling mechanism, is presented in Sections 4.2 to 4.4.

4.2. Conflict correction between different types of evidence

In order to measure the conflict arising between indirect and direct evidence, the concept of falsity is introduced. The falsity [58]

Table 1
Sources of trustworthiness evidence.

Phase	Sources of evidence
Planning Phase	Software Development Task Book Software Development Plan Software Configuration Management Plan Software Quality Assurance Plan
Design Phase	Software Requirements Specification Software Design Specification
Testing Phase	Software Test Plan Software Test Specification Software Test Report Software Quality Assurance Report
Delivery Phase	Software Release Notes Software User Manual Software Development Summary Report Software Configuration Management Report

is an index derived from the Shafer's conflict coefficient, designed to quantitatively describe the degree of conflict among different pieces of evidence and to reflect the negative impact of an individual evidence source on the overall evidential system. When applied to measure the conflict between indirect and direct evidence, it can accurately capture the extent to which indirect evidence interferes with direct evidence during fusion, thereby revealing additional conflict information introduced by the indirect source. Moreover, by computing falsity, it becomes possible to identify and appropriately down-weight highly conflicting evidence during the fusion process, leading to a more balanced allocation of weights. This method effectively suppresses the uncertainty and bias caused by conflicting indirect evidence, ultimately enhancing the stability and reliability of the fusion outcome.

In this paper, we categorize all evidence into direct and indirect sources, enabling a more precise characterization of the conflict between the two during the fusion process.

Let the set of direct evidence be denoted as:

$$M_d = \{m_d^1, m_d^2, \ldots, m_d^{J_d}\} \tag{2}$$

Let the set of indirect evidence be denoted as:

$$M_i = \{m_i^1, m_i^2, \ldots, m_i^{J_i}\} \tag{3}$$

Let the set of all evidence be denoted as:

$$M = M_d \cup M_i \tag{4}$$

where the total number of evidence is: $J = J_d + J_i$.

Using the binary conflict measurement function based on the Shafer's conflict coefficient (Eq. (A9)), we first compute the global conflict coefficient among all evidence sources over the frame of discernment θ:

$$k_0 = C(m_d^1, m_d^2, \ldots, m_d^{J_d}, m_i^1, m_i^2, \ldots, m_i^{J_i}) \tag{5}$$

For any evidence m (either direct or indirect), the global conflict coefficient among the remaining evidence after excluding m is defined as:

$$k(m) = C(M \setminus \{m\}) \tag{6}$$

Accordingly, the falsity of evidence m is defined as:

$$F(m) = \frac{k_0 - k(m)}{1 - k(m)} \tag{7}$$

where $F(m) \in [0, 1]$; when $F(m) = 0$, the evidence has no impact on the global conflict, and when F(m)=1, it indicates that the evidence is the main source of conflict. In particular, when only two pieces of evidence exist (e.g., m_1 and m_2), we have $F(m_1) = F(m_2) = k_0$.

To avoid assigning zero weights to any evidence in certain cases, we normalize the falsity of each evidence and obtain the normalized falsity weight as follows:

$$\mathrm{Fal}(m) = \frac{\exp(-F(m))}{\sum_{x \in M} \exp(-F(x))}, \quad 0 \le \mathrm{Fal}(m) \le 1,\ m \in M \tag{8}$$

Finally, the falsity vector of all evidence is represented as:

$$\mathrm{Fal} = \left[\mathrm{Fal}(m_d^1), \ldots, \mathrm{Fal}(m_d^{J_d}), \mathrm{Fal}(m_i^1), \ldots, \mathrm{Fal}(m_i^{J_i})\right] \tag{9}$$

By using falsity, the contribution of direct and indirect evidence to conflict correction can be quantified separately. In particular, by comparing the global conflict coefficient when both indirect and direct evidence are included versus when only direct evidence is used, one can precisely capture the extent to which indirect evidence interferes with direct evidence during the fusion process. The normalized falsity weights allow us to appropriately reduce the influence of highly conflicting indirect evidence, thereby improving the stability and reliability of the final assessment result.

To demonstrate the practical utility of the proposed falsity indicator $Fal(m)$ in assessing the conflict level of indirect evidence and controlling its fusion weight, we construct two illustrative scenarios that represent cases of high conflict and high consistency between indirect and direct evidence, respectively.

Let the frame of discernment be $\theta = \{A, B\}$, with its power set defined as $2^\theta = \{\emptyset, \{A\}, \{B\}, \{A, B\}\}$. Each body of evidence (BOE) has its basic belief assignment defined over this power set. In this example, we consider two direct evidences m_d^1 and m_d^2, and one indirect evidence m_i^1.

Case 1: High-conflict scenario

$$m_d^1(\{A\}) = 0.9, \quad m_d^1(\{A, B\}) = 0.1$$

$$m_d^2(\{A\}) = 0.85, \quad m_d^2(\{A, B\}) = 0.15$$

$$m_i^1(\{B\}) = 0.95, \quad m_i^1(\{A, B\}) = 0.05$$

The indirect evidence exhibits a strong belief focus toward proposition B, which is contrary to the preference expressed by the direct evidence. According to Eq. (5), the global conflict among the two direct pieces of evidence is $k_0^{DE} = 0.0288$. After introducing the indirect evidence, the overall conflict increases sharply to $k_0^{(All)} = 0.8970$. Based on Eq. (7), the falsity of m_i^1 is calculated as $Fal(m_i^1) = 0.8939$, indicating that this indirect evidence is the primary source of conflict. During the normalization stage (Eq. (8)), a relatively low weight is assigned to this evidence, effectively reducing its negative influence on the fusion result.

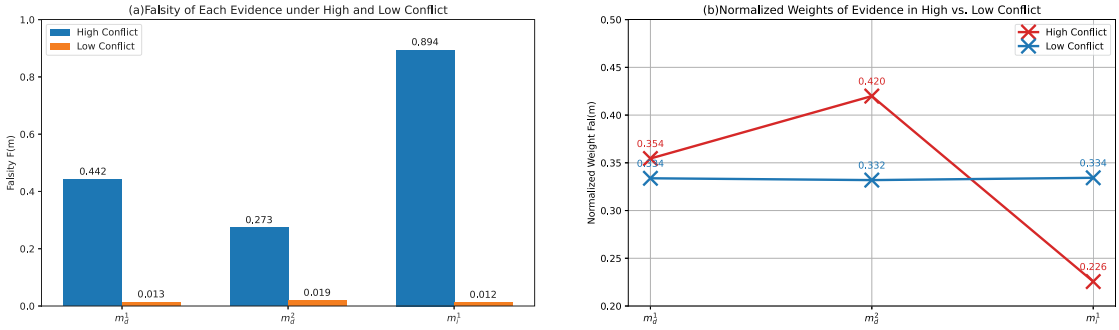
Case 2: Low-conflict scenario

$$m_d^1(\{A\}) = 0.9, \quad m_d^1(\{A, B\}) = 0.1$$

$$m_d^2(\{A\}) = 0.85, \quad m_d^2(\{A, B\}) = 0.15$$

$$m_i^1(\{A\}) = 0.88, \quad m_i^1(\{A, B\}) = 0.12$$

The indirect evidence remains highly consistent with the direct evidence, primarily supporting proposition A as well. According to Eq. (5), the global conflict among the two direct evidence is $k_0^{DE} = 0.0288$, and after introducing the indirect evidence, the overall conflict



(a) Falsity in high vs. low conflict

(b) Normalized weights in high vs. low conflict

Fig. 5. Comparison of Falsity and Normalized Weight Allocation Across Two Scenarios.

becomes $k_0^{All} = 0.0402$, showing no significant increase. Based on Eq. (7), the falsity ofm_i^1 is calculated as $Fal(m_i^1) = 0.0118$, indicating that the indirect evidence does not generate substantial conflict with the others. During the normalization stage (Eq. (8)), this evidence is assigned a weight roughly consistent with the others.

Fig. 5(a) compares the falsity values $Fal(m)$ of each piece of evidence (m_d^1, m_d^2, m_i^1)under two different scenarios. In the high-conflict scenario, the falsity of the indirect evidence m_i^1 reaches 0.89, significantly higher than that of the other evidence, while in the low-conflict scenario, all falsity values are close to zero, indicating strong overall consistency. Fig. 5(b) illustrates the normalized weight allocation across all evidence. In the high-conflict case, the model effectively suppresses the influence of the indirect evidence, thereby preventing it from skewing the assessment results. In the low-conflict case, the model retains all evidence fairly, which enhances the trustworthiness of the fusion outcome.

In summary, the proposed falsity-based method for handling indirect and direct evidence enables flexible adaptation under both high-conflict and low-conflict scenarios, providing more reliable inputs for subsequent fusion processes.

4.3. Conflict correction within the same type of evidence

After completing the first-stage conflict correction between indirect and direct evidence, this paper further addresses internal inconsistencies by conducting local consistency assessments within the indirect and direct evidence sets, respectively. Specifically, we introduce two indicators, uncertainty and credibility, to quantify both internal consistency and self-contradiction within each category of evidence, thereby enabling fine-grained weight adjustment. Uncertainty captures the inherent vagueness or ambiguity of evidence, helping to identify those instances where conclusions are difficult to draw due to insufficient information or high noise levels. Credibility, on the other hand, assesses the trustworthiness of evidence, effectively distinguishing high-quality sources from less reliable ones. In scenarios with significant internal conflicts, more credible evidence is granted greater influence during fusion. By combining these two indicators, the method reflects the overall reliability of evidence in supporting propositions. It not only accounts for the ambiguity of information but also emphasizes the reliability of the source. This dual perspective helps mitigate the negative impact of low-quality or ambiguous evidence on overall decision-making, thereby enhancing the accuracy and robustness of the final fusion outcome.

4.3.1. Credibility calculation

For each category of evidence (either indirect or direct), the pairwise Jousselme distances between all evidence elements are first calculated using Eq. (A6), forming a distance matrix. For any two pieces of evidence m_p and m_q within the same category $(1 \le p \le J, 1 \le q \le J)$, their Jousselme distance is given by:

$$d_{\mathrm{BBA}}(m_p, m_q) = \sqrt{\frac{1}{2}(m_p - m_q)^T D(m_p - m_q)}, \quad d_{\mathrm{BBA}}(m_p, m_q) \in [0,1] \tag{10}$$

Accordingly, the distance matrix for indirect evidence DMM_i and that for direct evidence DMM_d can be constructed as:

$$\mathrm{DMM} = \begin{bmatrix} 0 & d_{\mathrm{BBA}}(m_1, m_2) & \cdots & d_{\mathrm{BBA}}(m_1, m_J) \\ d_{\mathrm{BBA}}(m_2, m_1) & 0 & \cdots & d_{\mathrm{BBA}}(m_2, m_J) \\ \vdots & \vdots & \ddots & \vdots \\ d_{\mathrm{BBA}}(m_J, m_1) & d_{\mathrm{BBA}}(m_J, m_2) & \cdots & 0 \end{bmatrix} \tag{11}$$

These distance matrices are then transformed into similarity matrices as follows:

$$s_{pq} = 1 - d_{\mathrm{BBA}}(m_p, m_q), \quad 0 \le s_{pq} \le 1 \tag{12}$$

$$S = \begin{bmatrix} 1 & s_{12} & \cdots & s_{1J} \\ s_{21} & 1 & \cdots & s_{2J} \\ \vdots & \vdots & \ddots & \vdots \\ s_{J1} & s_{J2} & \cdots & 1 \end{bmatrix} \tag{13}$$

Thus, the similarity matrices for direct and indirect evidence are denoted as S_d and S_i, respectively. For each individual piece of evidence, its credibility is defined as follows:

$$\mathrm{Crd}(m_p) = \frac{\sum_{q=1, q\neq p}^{J} s_{pq}}{\sum_{l=1}^{J} \sum_{p=1, p\neq l}^{J} s_{pl}}, \quad 0 \le \mathrm{Crd}(m_p) \le 1, \quad m_p \in M_d \cup M_i \tag{14}$$

The credibility vectors for direct and indirect evidence are then obtained as follows:

$$\mathrm{Crd}_d = \left[\mathrm{Crd}_d^1, \mathrm{Crd}_d^2, \ldots, \mathrm{Crd}_d^J\right] \tag{15}$$

$$\mathrm{Crd}_i = \left[\mathrm{Crd}_i^1, \mathrm{Crd}_i^2, \ldots, \mathrm{Crd}_i^J\right] \tag{16}$$

The final credibility vector is expressed as:

$$\mathrm{Crd} = \mathrm{Crd}_d \parallel \mathrm{Crd}_i \tag{17}$$

4.3.2. Uncertainty calculation

To compute the uncertainty vector, we adopt the negation of evidence as the measurement approach. This method intuitively reflects the ambiguity and indeterminacy within the evidence. The negation of evidence, by "extracting information from the opposite", helps to capture and quantify complex and dynamic uncertainty. Especially in scenarios where system information continuously increases or changes, the information redistribution mechanism introduced by the negation operation enhances the model's flexibility and adaptability, thereby enabling a more accurate representation and handling of uncertainty embedded in the evidence.

For each piece of evidence m_j in a homogeneous evidence set ($j = 1, \ldots, J$), its negation $\overline{m}_j$ is computed according to Eqs. (A12)(A13)(A14). The original evidence m_j and its negation $\overline{m}_j$ are treated as two distinct sources of evidence. The conflict between these two sources is used to measure the uncertainty associated with m_j. This conflict is denoted as IV_j, and can be expressed as:

$$\mathrm{IV}_j = C(m, \overline{m}) \tag{18}$$

The more similar the evidence m_j is to its negation $\overline{m}_j$, the smaller the conflict degree IV_j, and the higher the uncertainty of m_j. The conflict measure IV_j is then normalized to derive the uncertainty Unc_j of evidence m_j, as follows:

$$\mathrm{Unc}_j = \frac{\mathrm{IV}_j}{\sum_{i=1}^{J} \mathrm{IV}_i}, \quad 1 \le j \le J \tag{19}$$

For each type of evidence, the uncertainty vectors are computed separately. Specifically, the uncertainty vectors for direct and indirect evidence are defined as follows:

$$\mathrm{Unc}_d = \left[\mathrm{Unc}_d^1, \mathrm{Unc}_d^2, \ldots, \mathrm{Unc}_d^J\right] \tag{20}$$

$$\mathrm{Unc}_i = \left[\mathrm{Unc}_i^1, \mathrm{Unc}_i^2, \ldots, \mathrm{Unc}_i^J\right] \tag{21}$$

The final uncertainty vector combining both direct and indirect evidence is given by:

$$\mathrm{Unc} = \mathrm{Unc}_d \parallel \mathrm{Unc}_i \tag{22}$$

To further validate the rationality and adaptability of the second-stage internal conflict resolution mechanism, this paper constructs two illustrative cases. These correspond to two representative scenarios: Case 1 involves significant inconsistency within indirect evidence, while Case 2 represents high consistency among direct evidence. The goal is to quantitatively assess whether the proposed indicators, credibility (Crd) and uncertainty (Unc), can effectively capture internal conflicts within each evidence set. Let the frame of discernment be $\theta = \{A, B\}$, and its power set be defined as $2^\theta = \{\emptyset, \{A\}, \{B\}, \{A, B\}\}$. Each BBA is defined over this power set.

Case 1: High-conflict scenario

$$m_i^1(\{A\}) = 0.9, \quad m_i^1(\{A, B\}) = 0.1$$

$$m_i^2(\{A\}) = 0.15, \quad m_i^2(\{A, B\}) = 0.85$$

$$m_i^3(\{B\}) = 0.95, \quad m_i^3(\{A, B\}) = 0.05$$

In this scenario, the three indirect pieces of evidence exhibit preference for different propositions, with significant structural differences, indicating an expected low level of internal consistency. Based on Eqs. (10) to (13), the computed Jousselme distance matrix and corresponding similarity matrix are as follows:

$$\mathrm{DMM}_i = \begin{bmatrix} 0 & 0.5303 & 0.9253 \\ 0.5303 & 0 & 0.6801 \\ 0.9253 & 0.6801 & 0 \end{bmatrix}$$

$$S_d = \begin{bmatrix} 1 & 0.4697 & 0.0747 \\ 0.4697 & 1 & 0.3199 \\ 0.0747 & 0.3199 & 1 \end{bmatrix}$$

The resulting credibility vector derived from the similarity matrix is:

$$\mathrm{Crd}_d = [0.3149,\ 0.4568,\ 0.2283]$$

Among the three pieces of indirect evidence, the second item m_i^2 exhibits relatively higher similarity with the other two, and thus obtains the highest credibility value (0.4568) after normalization. In contrast, the third piece of evidence m_i^3 shows very low similarity with the other two, only 0.0747 and 0.3199, resulting in the lowest credibility score (0.2283). This indicates that m_i^3 is structurally inconsistent with the rest of the evidence set.

Meanwhile, based on the conflict between each piece of evidence and its corresponding negation, the resulting uncertainty vector is obtained as follows:

$$\mathrm{Unc}_d = [0.3131,\ 0.3785,\ 0.3084]$$

Case 2: Low-conflict scenario

$$m_d^1(\{A\}) = 0.9, \quad m_d^1(\{A, B\}) = 0.1$$

$$m_d^2(\{A\}) = 0.85, \quad m_d^2(\{A, B\}) = 0.15$$

$$m_d^3(\{A\}) = 0.88, \quad m_d^3(\{A, B\}) = 0.12$$

The three direct pieces of evidence primarily support proposition A, with only minor perturbations observed. This indicates a high expected level of internal consistency and trustworthiness. Based on Eqs. (10) to (13), the calculated Jousselme distance matrix and the corresponding similarity matrix are as follows:

$$\mathrm{DMM}_d = \begin{bmatrix} 0 & 0.0354 & 0.0141 \\ 0.0354 & 0 & 0.0212 \\ 0.0141 & 0.0212 & 0 \end{bmatrix}$$

$$S_d = \begin{bmatrix} 1 & 0.9646 & 0.9859 \\ 0.9646 & 1 & 0.9788 \\ 0.9859 & 0.9788 & 1 \end{bmatrix}$$

The resulting credibility vector derived from the similarity matrix is:

$$\mathrm{Crd}_d = [0.3329,\ 0.3317,\ 0.3353]$$

The results show minimal variation in the credibility values among the three direct pieces of evidence, indicating stable performance that aligns with their expected structural consistency. Meanwhile, based on the conflict between each piece of evidence and its corresponding negation, the resulting uncertainty vector is calculated as follows:

$$\mathrm{Unc}_d = [0.3310,\ 0.3360,\ 0.3330]$$

The uncertainty values of all evidence pieces are very close, further indicating that this category of evidence possesses a clear information structure and high stability.

Fig. 6 illustrates the comparative curves of evidence Credibility (Crd) and Uncertainty (Unc) under two representative scenarios, while Fig. 7 presents the structural similarity between any two pieces of evidence within each type. The left panel corresponds to Case 1, where the indirect evidence exhibits significant structural differences, and the right panel shows Case 2, where the direct evidence maintains high internal consistency. In Case 1, the Crd curve fluctuates significantly, and the similarity matrix shows generally low values, indicating that the set of evidence suffers from structural ambiguity. More specifically, the first piece of evidence shows medium Crd (0.3149) and medium Unc (0.3131), which are nearly identical, reflecting its partly ambiguous nature without strong consistency or strong conflict. The second evidence, having the highest similarity with others, achieves the largest Crd (0.4568), where credibility clearly dominates over uncertainty (0.3785), indicating stronger reliability. In contrast, the third evidence, with very low similarity to the rest, yields the lowest Crd (0.2283) while its Unc (0.3084) exceeds credibility, revealing its poor alignment with the collective evidence set. This differentiation confirms that the proposed method can effectively highlight structural disparities among heterogeneous evidence in high-conflict situations. In contrast, Case 2 demonstrates consistently similar values of Crd and Unc across all three pieces of evidence. All credibility values converge around one third (0.3329, 0.3317, 0.3353), and uncertainty values are almost identical (0.3310, 0.3360, 0.3330). Together with pairwise similarities exceeding 0.96, this indicates that the direct evidence set

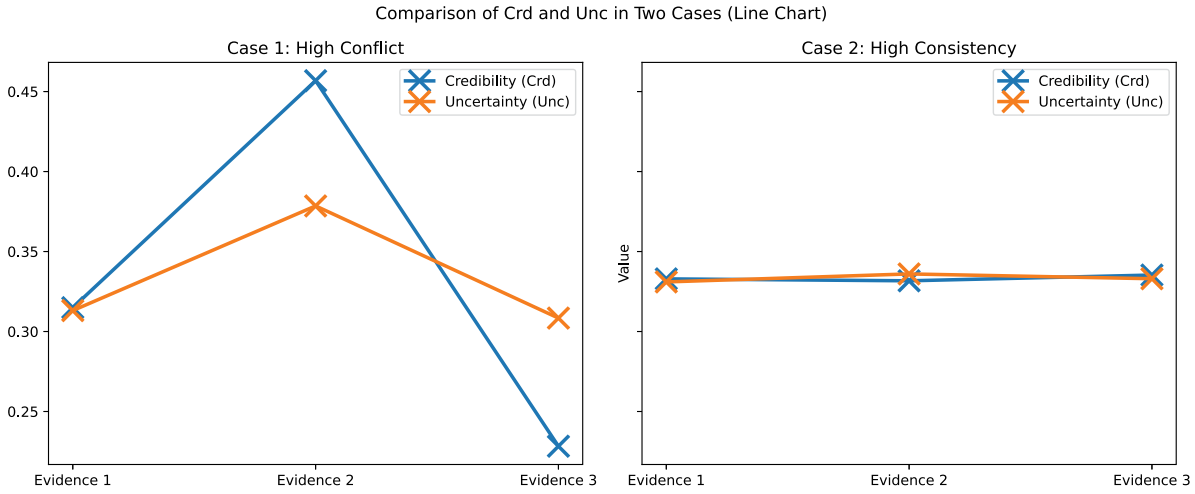


Fig. 6. Comparison of Evidence Credibility (Crd) and Uncertainty (Unc) Curves.

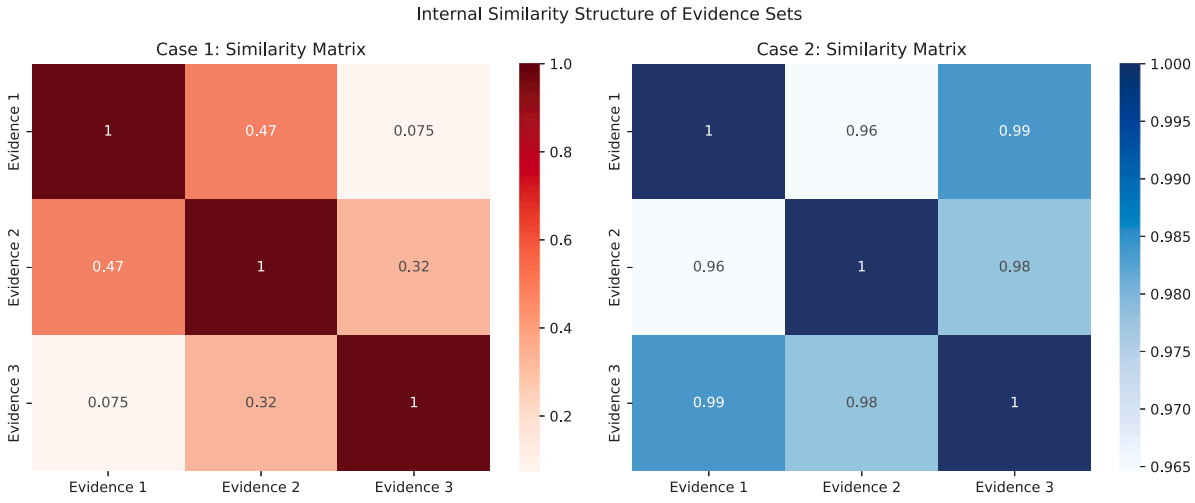


Fig. 7. Similarity Matrices in Two Scenarios.

is highly homogeneous and well-defined. The contrast between Case 1 and Case 2 demonstrates that the proposed approach behaves as expected: it differentiates evidence appropriately under high conflict and remains stable under low conflict, thereby verifying the rationality of the algorithm.

4.4. Fusion of multiple types of evidence

To resolve conflicts among pieces of evidence, it is necessary to generate a composite discounting factor to correct the original evidence values. Different assessment metrics, credibility, falsity, and uncertainty, produce distinct vectors that reflect the characteristics of evidence from different dimensions. Although these vectors vary in nature, each plays a positive role in adjusting the reliability of the evidence. To ensure the comprehensiveness of the final discounting factor, it is essential to preserve as much information from these vectors as possible. This helps reconcile discrepancies among the metrics and enhances the stability and precision of evidence adjustment. To achieve this, the proposed method uses the computed falsity, credibility, and uncertainty vectors as the base weights for each piece of evidence. A weighted combination strategy inspired by game theory [59] is then employed to derive an optimal integrated weight, referred to as the composite discounting factor, which supports scientifically grounded correction of evidence uncertainty.

First, a weighting model is constructed. Based on the falsity vector Fal, the credibility vector Crd, and the uncertainty vector Unc, the basic set of weight vectors is defined as:

$$W = \{\varphi_k \mid k = 1, 2, 3\} \tag{23}$$

$$\varphi_1 = \text{Fal}, \quad \varphi_2 = \text{Crd}, \quad \varphi_3 = \text{Unc} \tag{24}$$

Any linear combination of these three vectors can be expressed as:

$$\phi = \sum_{k=1}^{3} \lambda_k \varphi_k^T, \quad \lambda_k > 0, \quad \sum_{k=1}^{3} \lambda_k = 1 \tag{25}$$

where ϕ is the synthesized weight vector, and λ_k denotes the linear combination coefficient associated with each individual component φ_k.

Next, a system of linear equations is established. Based on the game-theoretic weighted combination strategy, the goal is to optimize the linear combination coefficients $\lambda_k (k = 1, 2, 3)$ such that the deviation between the synthesized weight vector φ and each original weight vector φ_k is minimized, i.e.,

$$\min \frac{1}{2} \|\varphi - \varphi_k\|_2^2 \tag{26}$$

By applying the principle of differentiation to derive Eq. (25), the following linear system of equations is obtained:

$$\begin{bmatrix} \varphi_1^T\varphi_1 & \varphi_2^T\varphi_1 & \varphi_3^T\varphi_1 \\ \varphi_1^T\varphi_2 & \varphi_2^T\varphi_2 & \varphi_3^T\varphi_2 \\ \varphi_1^T\varphi_3 & \varphi_2^T\varphi_3 & \varphi_3^T\varphi_3 \end{bmatrix} \cdot \begin{bmatrix} \lambda_i^1 \\ \lambda_i^2 \\ \lambda_i^3 \end{bmatrix} = \begin{bmatrix} \varphi_1^T\varphi_1 \\ \varphi_2^T\varphi_2 \\ \varphi_3^T\varphi_3 \end{bmatrix} \tag{27}$$

Then, the composite discounting factors are calculated. According to Eq. (26), the linear combination coefficients $\lambda_k (k = 1, 2, 3)$ are first obtained and then normalized as follows:

$$\lambda_k^* = \frac{|\lambda_k|}{\sum_{k=1}^{3} |\lambda_k|} \tag{28}$$

According to Eq. (27), the optimal integrated weight vector is given by:

$$\varphi^* = \sum_{k=1}^{3} \lambda_k^* \varphi_k^T \tag{29}$$

$$\varphi^* = [\varphi_1^*, \varphi_2^*, \ldots, \varphi_J^*] \tag{30}$$

where $\varphi_1^* \ldots \varphi_J^*$ denote the final composite discounting factors corresponding to each evidence $m_1 \ldots m_J$, respectively. These factors are used to adjust the evidential support of each source prior to fusion.

Finally, each piece of evidence m_j is revised using its corresponding synthesized discount factor $\varphi_j^* (j = 1, \ldots, J)$, resulting in the weighted average evidence $\tilde{m}$ as follows:

$$\tilde{m} = \sum_{j=1}^{J} \left(\varphi_j^* \times m_j \right), \quad 1 \le j \le J \tag{31}$$

Given that there are J pieces of evidence, the weighted average evidence $\tilde{m}$ is sequentially combined $J - 1$ times using Dempster's combination rule to obtain the final fusion result:

$$F(\tilde{m}) = ((((\tilde{m} \oplus \tilde{m})_1 \oplus \tilde{m})_2 \oplus \ldots)_{J-2} \oplus \tilde{m})_{J-1} \tag{32}$$

where $F(\tilde{m})$ represents the final fusion result.

Algorithm 1 A Multi-Stage Evidence Fusion Algorithm

Require:
$M_{evidence} = \{m_1, m_2, \ldots, m_J\}$
$u(F) = \{u(F_0), \ldots, u(F_5)\}$
$F = F_0, \ldots, F_5$
Ensure: $\tilde{\mu}_{subAttr}$

Step 1: Evidence utility transformation
for each evidence m_j in $M_{evidence}$ **do**
  if m_j is qualitative **then**
    $u_j \leftarrow \sum \lambda_n \cdot \mu(F_n)$
  else if e_j is quantitative **then**
    $u_j \leftarrow$ interpolate using Eq. (A2)
  end if
  $BBA_j \leftarrow$convert $u_j$to mass function via Eq. (A3)
end for
Step 2: Multi-stage evidence conflict mitigation
Split $M_evidence$ into M_d (direct) and M_i (indirect)
Compute falsity $F(m_j), \quad \forall m_j \in M_d \cup M_i$
Compute credibility $\mathrm{Crd}_j, \quad \forall m_j \in M_d \cup M_i$
Compute uncertainty $\mathrm{Unc}_j, \quad \forall m_j \in M_d \cup M_i$
Combine $[Fal, Crd, Unc]$ to get φ_j^* via game-theoretic weighting
Step 3: Discount and average the corrected evidence
for each $m_j \in M_{evidence}$ **do**
  $\tilde{m}_j \leftarrow \varphi_j^* \cdot m_j$
end for
Step 4: Evidence fusion
$\tilde{m} \leftarrow \tilde{m}_1 \oplus \tilde{m}_2 \oplus \ldots \oplus \tilde{m}_J$
Step 5: Utility reverse transformation
$\tilde{\mu}_{subAttr} \leftarrow u(\tilde{m})$ using Eq. (A4)
return $\tilde{\mu}_{subAttr}$

The complete fusion process of trustworthy evidence is summarized in Algorithm 1. Based on the unified utility representation, the algorithm begins by mapping raw qualitative or quantitative indicators into utility values. These utility values are then used to construct BBAs according to predefined trust levels specified by domain experts. Next, the algorithm identifies and resolves evidence conflicts through a multi-stage, type-specific adjustment strategy. First, it computes the falsity to quantify the structural divergence between direct and indirect evidence. Then, it calculates credibility and uncertainty vectors using evidence similarity and negation-based opposition, respectively, to characterize the internal consistency and clarity of information within each evidence group. Subsequently, a weighted combination method based on game theory is applied to integrate the three dimensions of falsity, credibility, and uncertainty into a composite discount factor. These factors are used to dynamically correct and reweight the original evidence. Finally, all adjusted evidence is fused using the Dempster's rule of combination, and the resulting mass function is mapped back to utility values to obtain the final trustworthiness assessment of the sub-attribute.

5. Empirical study

5.1. Experimental design

To validate the rationality, effectiveness, and superiority of the proposed trustworthiness assessment model based on the classification of direct and indirect evidence, this paper designs a systematic experimental framework. The experiments are structured around three key aspects: verification of theoretical properties, comparison of assessment accuracy, and assessment of the evidence fusion performance. Specifically, the analysis addresses the following three research questions:

RQ 1: Does the proposed software trustworthiness assessment model satisfy the fundamental criteria for trustworthiness measurement?

To verify the theoretical soundness of the proposed model, we examine it against the following four fundamental measurement criteria, which have been validated for their applicability and effectiveness in Ref. [60].:

- Monotonicity: As the values of trustworthy evidence increase, the corresponding attribute value should increase monotonically;
- Acceleration: When only one evidence value increases while others remain unchanged, its marginal contribution to the attribute value should gradually diminish;
- Sensitivity: Relative changes in trustworthy evidence should be quantitatively reflected in the corresponding changes of the attribute value;
- Substitutability: When the overall attribute value remains constant, changes in two evidence values should trigger a dynamic adjustment in their mutual influence, reflecting a substitutive relationship between the evidence sources.

RQ 2: Does the proposed model improve the accuracy of trustworthiness estimation compared with assessment methods that do not distinguish between evidence types?

This research question aims to verify whether the introduction of the "evidence type classification + conflict resolution + weighted discount" mechanism enables the proposed model to more accurately and precisely reflect the actual level of software trustworthiness, and whether it yields assessment results that are closer to the expected values compared with traditional models.

RQ 3: Does the proposed multi-stage conflict-aware evidence fusion method improve the accuracy and robustness of the fusion results compared with existing state-of-the-art evidence fusion algorithms?

To validate the effectiveness of the proposed fusion strategy, several state-of-the-art methods are selected for comparison. The performance of the proposed method is assessed under complex conditions such as high conflict, elevated uncertainty, and bias interference.

Table 2
Assessability of trustworthiness attributes for a certain software system.

Software system	Trustworthiness attribute	Number of sub-attributes	Number of evidence	Assessable or not
A certain software system	Reliability	**4**	**21**	✓
	Security	6	18	✓
	Functionality	2	7	✓
	Efficiency	3	15	✓
	Compatibility	2	9	×
	Usability	5	26	×
	Maintainability	6	22	✓
	Portability	3	18	×

5.2. Data collection and preparation

To ensure the scientific rigor and trustworthiness of the assessment results, the experimental data used in this paper were collected from a real-world system following the principle of triangulation. Three complementary sources were utilized: (1) Development process traceability data, obtained from life-cycle documentation provided by the software developer, including requirements specifications, design documents, and configuration records; (2) Standardized laboratory test data, generated by structured testing conducted by a assessment agency; and (3) Environmental simulation data, acquired through operational trials on a high-fidelity simulation platform.

A hierarchical trustworthiness assessment model based on direct and indirect evidence is adopted to assess software trustworthiness. As presented in Table 2, the assessment of the software involves multiple attributes, with a consistent fusion strategy applied across them. To avoid redundant elaboration, reliability is selected in this section as a representative attribute for detailed assessment and demonstration. The corresponding indicators and their evidence values are shown in Table 3. The assessment model for this attribute is formulated in Eq. (33).

$$R = \alpha M \oplus \beta U \oplus \gamma R \oplus \delta F \tag{33}$$

In Eq. (33), R denotes the overall reliability assessment score, which is composed of four dimensions: Maturity (M), Usability (U), Recoverability (R), and Fault Tolerance (F). The coefficients α to δ represent composite discounting factors obtained through a game-theoretic weighted combination. The operator $\oplus$ denotes the Dempster rule-based evidence fusion. Additionally, the bodies of evidence (BOE) are defined as follows: indirect evidence is denoted as indirectm, while direct evidence is represented as m.

In addition, to ensure the objectivity and reliability of accuracy assessment for the proposed trustworthiness estimation method, a ground truth value was assigned to each sub-attribute in the experiment, serving as a benchmark for comparing the outputs of different fusion methods. These ground truth values were determined based on the following principles:

•Expert consensus: For each sub-attribute, domain experts were invited to assess the trustworthiness level based on the credibility of the evidence sources, content relevance, and data quality. Through multiple rounds of feedback aggregation, a consensus-based baseline estimation was established.

•Stable simulation outputs: Under an ideal simulation environment without interference from evidence conflicts or uncertainty factors, multiple rounds of the fusion process were conducted. After statistical analysis, the output showed a clear convergence trend, and the stabilized estimation was used to further refine the ground truth.

Through the integration and validation of the above multi-source information, the established ground truth serves as a high-confidence reference benchmark for trustworthiness assessment at the sub-attribute level.

Before performing evidence fusion on the basic indicators, it is necessary to unify the dimensional scales of the measurement data to ensure comparability across different indicators. Therefore, we adopt a utility-based transformation approach (Appendix A.1) to map the original indicator values into a standardized utility space. To this end, we introduce a trusted evaluation utility scale, which is defined by expert judgment and consists of five levels $H_1 \sim H_5$. Their corresponding utility values increase uniformly over the interval $[0, 1]$, as follows:

$$\theta = [H_1 : 0.00, H_2 : 0.25, H_3 : 0.5, H_4 : 0.75, H_5 : 1.00]$$

Based on this recognition framework θ, we use Eqs. (A2) and (A3) to convert the quantitative values of indirect evidence and direct evidence into Basic Belief Assignments (BBA). The converted BBA values represent the distribution of evidence over different trust levels, thereby providing the basis for subsequent fusion computations.

As an illustrative example, consider the evidence "Fault Repair Rate".

(1) Interval Determination The utility scale of the recognition framework is defined as:

$$\mathrm{u}(H_1) = 0.00, u(H_2) = 0.25, u(H_3) = 0.50, u(H_4) = 0.75, u(H_5) = 1.00$$

The original measurement value is $v_\beta = 0.66$, which falls within the interval $[u(H_3), u(H_4)] = [0.50, 0.75]$. Therefore, the interpolation and assignment are performed using these two adjacent levels.

(2) Computation of Utility Value

According to Eq. (A2), the interpolated utility is:

$$\mathrm{u}(v_\beta) = u(H_4) - \frac{u(H_4) - v_\beta}{u(H_4) - u(H_3)}(u(H_4) - u(H_3))$$

Substituting values:

$$\mathrm{u}(v_\beta) = 0.75 - \frac{0.75 - 0.66}{0.75 - 0.50} \times (0.75 - 0.50) = 0.66$$

Thus, the utility corresponding to the raw measurement is $u(v_\beta) = 0.66$

(3) Deriving the BBA

The interpolated utility is then allocated between the adjacent levels H_3 and H_4 according to Eq. (A3):

$$\mathrm{m}(H_3) = \frac{u(H_4) - u(v_\beta)}{u(H_4) - u(H_3)} = 0.36$$

$$\mathrm{m}(H_4) = 1 - m(H_3) = 0.64$$

Table 4 presents the evidence values for different attributes and sub-attributes, along with their corresponding BBA distributions after transformation into the utility space.

Through the above transformation, the measurement indicators are normalized within a unified utility framework, ensuring their comparability and integrability, and thereby laying a solid foundation for subsequent trustworthiness evaluation.

5.3. RQ1: Compliance of the proposed software trustworthiness model with fundamental measurement principles

To verify whether the proposed trustworthiness assessment model satisfies the fundamental measurement principles of Monotonicity and Acceleration, this experiment takes the sub-attribute maturity as an

Table 3
Reliability-related indicators and corresponding evidence values of a given system.

Attribute	Sub-Attribute	EType	EName	Performance
Reliability	Maturity Ground Truth = 0.74	Indirect Evidence	Test Coverage	0.95
			Fault Repair Rate	0.66
			Test Stability	0.87
			Verification Pass Rate	0.84
		Direct Evidence	Fault Removal Rate	0.72
			Mean Fault Interval Satisfaction	0.68
			Fault Density Compliance	0.56
			Fault Retention Satisfaction	0.45
	Usability Ground Truth = 0.79	Direct Evidence	System Availability	0.90
			Uptime Under Special Conditions	0.67
	Recoverability Ground Truth = 0.75	Direct Evidence	Mean Recovery Time Satisfaction	0.75
			Backup Data Rate	0.77
			Recovery Originality	0.66
		Indirect Evidence	Accuracy of Recovery Requirement Design	0.95
			Adequacy of Mean Recovery Time Design	0.76
			Adequacy of Max Recovery Time Design	0.85
			Recovery Matching Consistency	0.85
	Fault tolerance Ground Truth = 0.64	Indirect Evidence	Redundancy of Key Components	0.76
			Implementation of Fault Handling Design	0.79
		Direct Evidence	Avoidance Failure Rate Satisfaction	0.45
			Anti-Failure Operation Compliance	0.55

Table 4
Mapping of indicators in the utility space.

Sub-Attribute	EType	EName	Performance	BBA
Maturity Ground Truth = 0.74	Indirect Evidence	Test Coverage	0.95	(‘H4’,): 0.2, (‘H5’,): 0.8
		Fault Repair Rate	0.66	(‘H3’,): 0.36, (‘H4’,): 0.64
		Test Stability	0.87	(‘H4’,): 0.52, (‘H5’,): 0.48
		Verification Pass Rate	0.84	(‘H4’,): 0.64, (‘H5’,): 0.36
	Direct Evidence	Fault Removal Rate	0.72	(‘H3’,): 0.12, (‘H4’,): 0.88
		Mean Fault Interval Satisfaction	0.68	(‘H3’,): 0.28, (‘H4’,): 0.72
		Fault Density Compliance	0.56	(‘H3’,): 0.76, (‘H4’,): 0.24
		Fault Retention Satisfaction	0.45	(‘H2’,): 0.2, (‘H3’,): 0.8
Usability Ground Truth = 0.79	Direct Evidence	System Availability	0.90	(‘H4’,): 0.4, (‘H5’,): 0.6
		Uptime Under Special Conditions	0.67	(‘H3’,): 0.32, (‘H4’,): 0.68
Recoverability Ground Truth = 0.75	Direct Evidence	Mean Recovery Time Satisfaction	0.75	(‘H4’,): 1.0
		Backup Data Rate	0.77	(‘H4’,): 0.92, (‘H5’,): 0.08
		Recovery Originality	0.66	(‘H3’,): 0.36, (‘H4’,): 0.64
	Indirect Evidence	Accuracy of Recovery Requirement Design	0.95	(‘H4’,): 0.2, (‘H5’,): 0.8
		Adequacy of Mean Recovery Time Design	0.76	(‘H4’,): 0.96, (‘H5’,): 0.04
		Adequacy of Max Recovery Time Design	0.85	(‘H4’,): 0.6, (‘H5’,): 0.4
		Recovery Matching Consistency	0.85	(‘H4’,): 0.6, (‘H5’,): 0.4
Fault tolerance Ground Truth = 0.64	Indirect Evidence	Redundancy of Key Components	0.76	(‘H4’,): 0.96, (‘H5’,): 0.04
		Implementation of Fault Handling Design	0.79	(‘H4’,): 0.84, (‘H5’,): 0.16
	Direct Evidence	Avoidance Failure Rate Satisfaction	0.45	(‘H2’,): 0.2, (‘H3’,): 0.8
		Anti-Failure Operation Compliance	0.55	(‘H3’,): 0.8, (‘H4’,): 0.2

example and selects one of its supporting evidence, m_4, as the variable, while keeping all other evidence constant. The original value of m_4 is gradually increased from 0.45 to 0.75, and the updated value is fed into the trustworthiness assessment model at each step. The resulting sub-attribute trustworthiness score $\tilde{\mu}_{subAttr}$ is recorded after each update. The incremental change in trustworthiness $\Delta\mu$ between consecutive updates is also calculated to analyze the behavior of Acceleration. As shown in Fig. 8, panel (a) illustrates that as the individual evidence value m_4 increases, the overall fused result $\tilde{\mu}_{subAttr}$ exhibits a clear monotonic upward trend with a smooth curve. This confirms the model's consistency and predictability under increasing input, satisfying the requirement of monotonicity. Panel (b) presents the increment $\Delta\mu$ in trustworthiness resulting from consecutive changes in m_4, which gradually decreases from 0.001 to 0.0003. This reflects a typical diminishing marginal effect. The result indicates that although the contribution of a particular piece of evidence increases, its marginal impact is gradually reduced. This prevents the model from over-amplifying high-value evidence, demonstrating a well-functioning acceleration mechanism.

To further verify whether the proposed trustworthiness assessment model satisfies the fundamental measurement principles of sensitivity and substitutability, two targeted experiments were conducted. For sensitivity, all sub-attribute evidence values were held constant except for m_1, which was perturbed by ±20% and ±10% from its original value of 0.72. The resulting changes in the fused outcome $\tilde{\mu}_{subAttr}$ were observed to assess whether the model responds significantly and appropriately to minor input variations. For substitutability, the total value of $m_1 + m_2$ was fixed at 1.4, and adjustments were made by increasing m_1 and decreasing m_2 accordingly. This test aimed to examine whether the overall trustworthiness remains relatively stable and whether the model can dynamically redistribute weights between evidences to achieve a compensatory substitution effect.

As shown in Fig. 9(a), when the value of m_1 is perturbed within the range of −20% to +20%, the overall fused trustworthiness exhibits

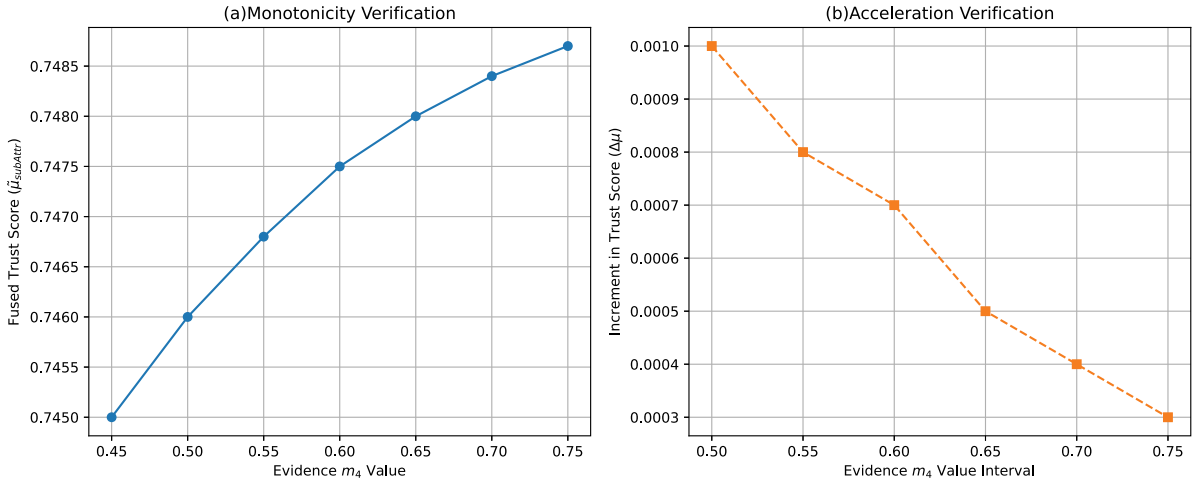


Fig. 8. Verification of Monotonicity and Acceleration.

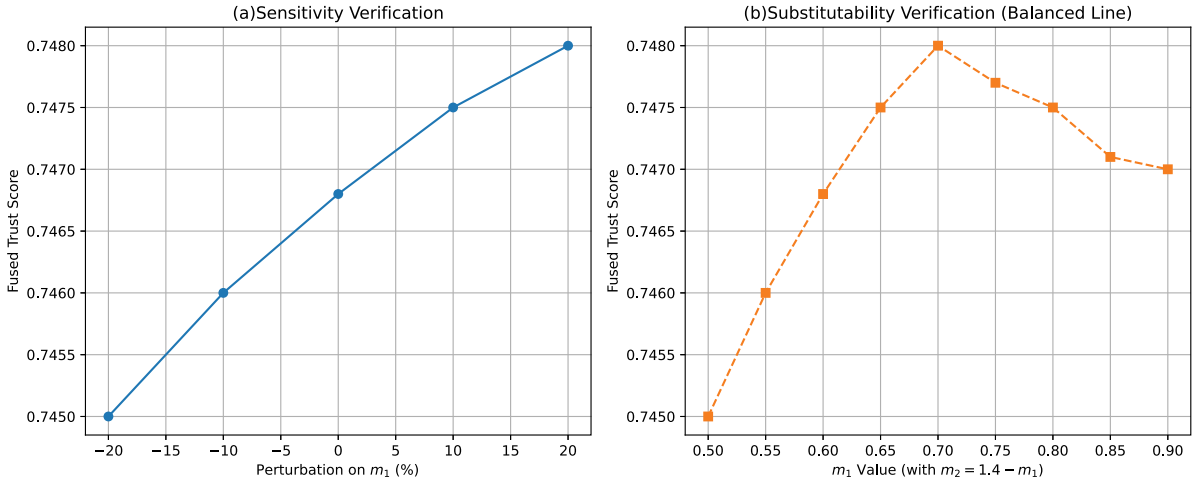


Fig. 9. Verification of Sensitivity and Substitutability.

an approximately linear increasing trend. The model responds more sensitively within the ±10% range, verifying its responsiveness and continuity in mapping small evidence variations, which is an essential requirement for sensitivity in trustworthiness assessment models. In Fig. 9(b), where m_1 increases and m_2 decreases accordingly, the resulting trustworthiness remains within a narrow fluctuation range of [0.745, 0.747], despite notable changes in the individual inputs. This indicates the presence of an internal compensatory mechanism within the model, demonstrating strong adaptability to substitutability.

5.4. RQ2: Performance of the assessment model with evidence type distinction

To examine whether the evidence type distinction strategy improves the accuracy of trustworthiness assessment, four key sub-attributes of reliability, including maturity, usability, recoverability, and fault tolerance, were chosen for comparative experiments As shown in Table 5 the results obtained from the proposed model are compared with those from a traditional undifferentiated fusion model (Baseline) [61], focusing on their deviations from the expected Ground Truth values. In terms of assessment metrics, three types of indicators are used to quantitatively assess the performance of the methods:

•Absolute Error: Measures the numerical deviation between the fused result and the ground truth:

$$\text{Error} = \left|\mu_{\text{model}} - \mu_{\text{true}}\right|$$

•Mean Absolute Error(MAE): Represents the average of absolute errors across all sub-attributes, reflecting overall accuracy:

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N}\left|\mu_i^{\text{model}} - \mu_i^{\text{true}}\right|$$

•Improvement Rate: Indicates the relative reduction in error achieved by the proposed method compared to the baseline:

$$\text{Rate} = \frac{\text{Error}_{\text{baseline}} - \text{Error}_{\text{proposed}}}{\text{Error}_{\text{baseline}}} \times 100\%$$

As illustrated in Fig. 10(a), the performance of both methods is compared across four sub-attributes, detailing the absolute error, error variation trends, improvement rates, and alignment with the ground truth. The Proposed method consistently exhibits lower absolute errors across all sub-attributes. Notably, in the fault tolerance dimension, the Baseline method yields an error of 0.013, whereas the Proposed method reduces this to only 0.003, achieving a reduction of over 75%. Fig. 10(b) shows that the Baseline method exhibits considerable fluctuations in recoverability and fault tolerance, while the Proposed

Table 5
Comparison of type-based and non-type-based models.

Sub-Attribute	Ground truth	Baseline output	Proposed output	Baseline error	Proposed error	Improvement rate (%)
Maturity	0.74	0.74603	0.74494	0.00603	0.00494	18.03809
Usability	0.79	0.78511	0.78511	0.00489	0.00489	0
Recoverability	0.75	0.75720	0.74505	0.00720	0.00495	43.67495
Fault tolerance	0.64	0.62714	0.64335	0.01286	0.00335	73.81901
MAE				**0.00774**	**0.00430**	**44.36609**

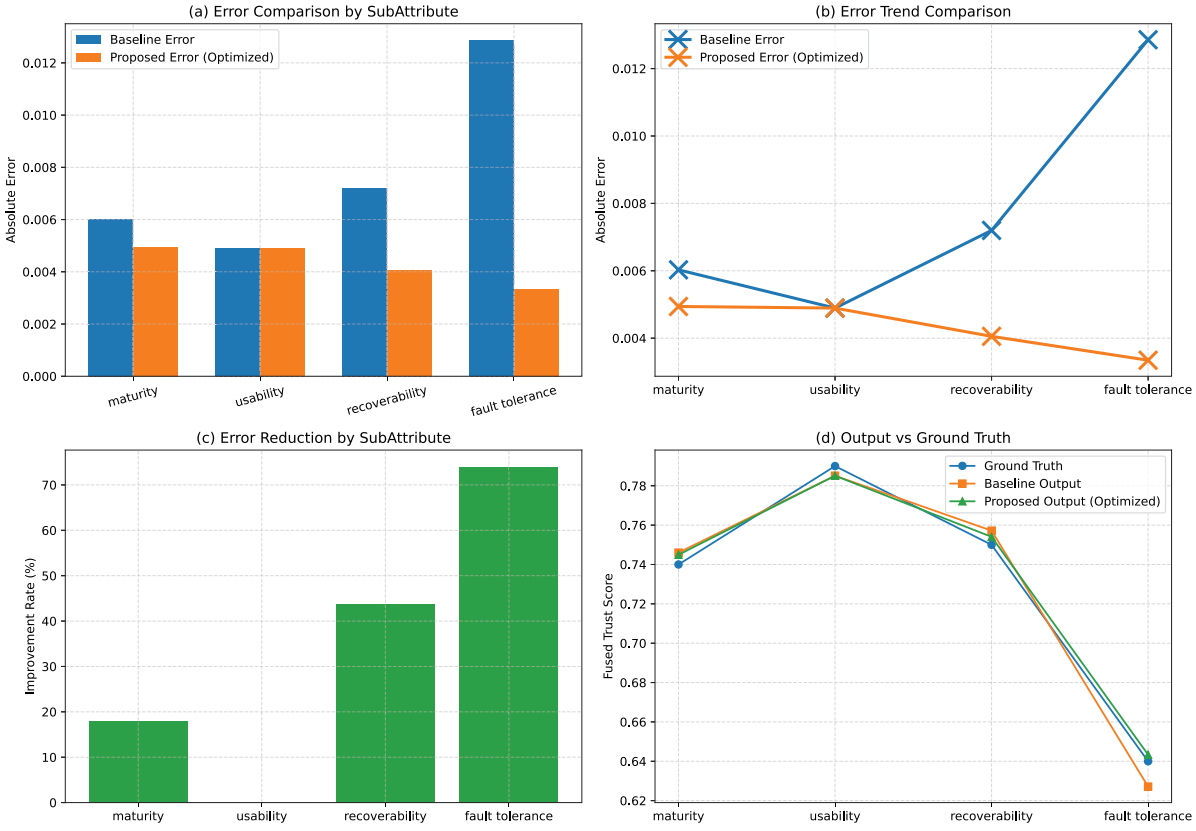


Fig. 10. Assessment Accuracy Comparison Between Type-Distinguishing and Non-Distinguishing Evidence Models.

method maintains a smoother curve with more stable and lower error values, demonstrating superior generalization capability.

In Fig. 10(c), all sub-attributes demonstrate positive improvement rates. Improvements in fault tolerance (74.4%) and recoverability (43.67%) are particularly significant. For usability, no improvement is observed due to the absence of indirect evidence, which indicates that the method is capable of identifying and attenuating inconsistent or low-quality evidence sources, thereby enhancing the reliability of fusion results. Fig. 10(d) further supports these findings. For all sub-attributes, the proposed method outputs are visibly closer to the Ground Truth curve, indicating a stronger fit and a more accurate estimation. The advantage is especially notable in the fault tolerance and usability dimensions, where the proposed method significantly outperforms the baseline.

Overall, this experiment confirms that incorporating evidence type distinction into the trustworthiness assessment process effectively improves the accuracy and robustness of fusion results.

5.5. RQ3: Performance of the multi-stage evidence fusion method

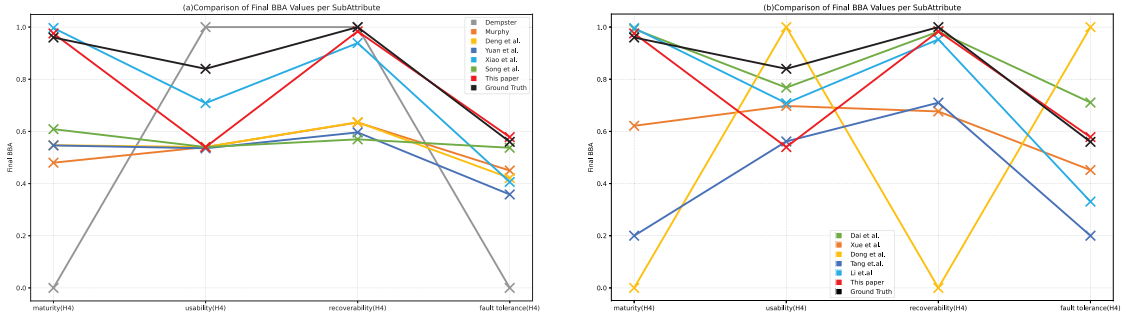
To further assess the performance of the proposed multi-stage evidence fusion method in the task of trustworthy evidence aggregation, this section investigates its effectiveness in improving both accuracy and robustness. The focus is placed on each method's ability to handle and correct conflicting evidence. The experiment continues to use evidence values associated with reliability-related sub-attributes, under a predefined frame of discernment $\theta = \{H_1 : 0.0, H_2 : 0.25, H_3 : 0.5, H_4 : 0.75, H_5 : 1.0\}$.

Three assessment metrics are adopted to quantitatively assess fusion performance: Mean Absolute Error (MAE), Absolute Error (AE), and Maximum Error. To ensure the representativeness and timeliness of the conclusions, this study compares the results with both classical evidence fusion algorithms and the latest evidence theory methods proposed in recent years. The fusion results of different methods are presented in Table 6, where the final outputs for the four sub-attributes are compared against the Ground Truth values. The results clearly demonstrate the assessment deviations among methods. The proposed method consistently achieves outputs closest to the Ground Truth across nearly all attributes, while the traditional Dempster method fails in high-conflict scenarios.

As illustrated in Fig. 11 (a), this study evaluates the performance of several classical evidence fusion methods against the proposed method in terms of the alignment between output values and ground truth on H4 (Target Credibility Level). The results indicate that, except for a slight deviation in the Usability attribute, the proposed method almost completely overlaps with the ground truth across the entire curve, demonstrating superior fitting capability and overall stability. In contrast, the Dempster method suffers from performance degradation

Table 6
Type-based comparison of BBA and final mass across sub-attributes.

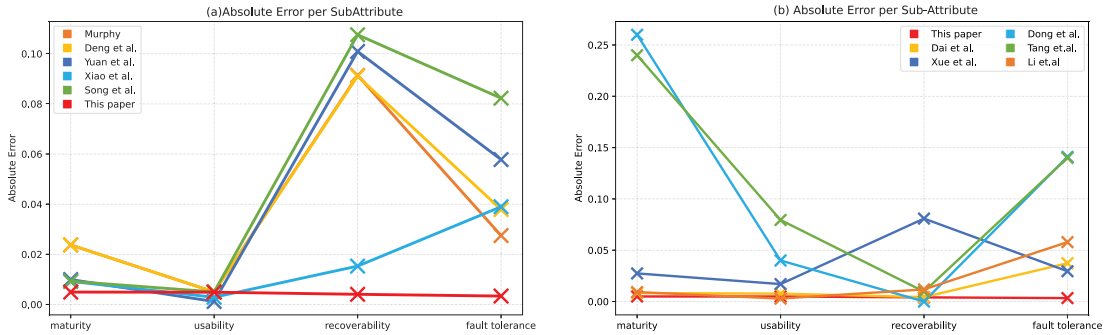
Method	Maturity		Usability		Recoverability		Fault tolerance	
	BBA	Fusion result	BBA	Fusion result	BBA	Fusion result	BBA	Fusion result
Dempster [51]	**conflict**	0	(‘H4’,): 1.0	0.7500	(‘H4’,): 1.0	0.7500	**conflict**	0
Murphy [62]	(‘H2’,): 0.0250 (‘H3’,): 0.2900 (‘H4’,): 0.4800 (‘H5’,): 0.2050	0.7162	(‘H4’,): 0.5400 (‘H3’,): 0.1600 (‘H5’,): 0.3000	0.7850	(‘H5’,): 0.3657 (‘H4’,): 0.6343	0.8414	(‘H5’,): 0.0500 (‘H3’,): 0.4000 (‘H2’,): 0.1000 (‘H4’,): 0.4500	0.6125
Deng et al.[63]	(‘H2’,): 0.0122 (‘H4’,): 0.5472 (‘H3’,): 0.2575 (‘H5’,): 0.1831	0.7163	(‘H4’,): 0.5400 (‘H3’,): 0.1600 (‘H5’,): 0.3000	0.7850	(‘H4’,): 0.6354 (‘H5’,): 0.3646	0.8411	(‘H4’,): 0.4213 (‘H3’,): 0.4255 (‘H5’,): 0.0468 (‘H2’,): 0.1064	0.6021
Yuan et al.[64]	(‘H3’,): 0.2504 (‘H4’,): 0.5462 (‘H2’,): 0.0109 (‘H5’,): 0.1925	0.7300	(‘H4’,): 0.5353 (‘H5’,): 0.3100 (‘H3’,): 0.1547	0.7888	(‘H4’,): 0.5962 (‘H5’,): 0.4038	0.8509	(‘H2’,): 0.1194 (‘H3’,): 0.4774 (‘H4’,): 0.3582 (‘H5’,): 0.0450	0.5822
Xiao et al.[16]	(‘H3’): 0.0033 (‘H4’): 0.9965 (‘H5’): 0.0001	0.7492	(‘H3’): 0.0601 (‘H4’): 0.7083 (‘H5’): 0.2316	0.7929	(‘H4’): 0.9388 (‘H5’): 0.0612	0.7653	(‘H2’): 0.0023 (‘H3’): 0.5914 (‘H4’): 0.4062 (‘H5’): 0.0001	0.6010
Song et al.[65]	(‘H3’,): 0.1972 (‘H4’,): 0.6086 (‘H5’,): 0.1942	0.7493	(‘H3’,): 0.1600 (‘H4’,): 0.5400 (‘H5’,): 0.3000	0.7850	(‘H4’,): 0.5696 (‘H5’,): 0.4304	0.8576	(‘H2’,): 0.1343 (‘H3’,): 0.5372 (‘H4’,): 0.2918 (‘H5’,): 0.0367	0.5577
Dai et al.[66]	(‘H3’,): 0.0072 (‘H4’,): 0.9925 (‘H5’,): 0.0003	0.7483	(‘H3’,): 0.0514 (‘H4’,): 0.7678 (‘H5’,): 0.1808	0.7823	(‘H4’,): 0.9827 (‘H5’,): 0.0173	0.7543	(‘H2’,): 0.0015 (‘H3’,): 0.2876 (‘H4’,): 0.7108 (‘H5’,): 0.0001	0.6774
Xue et al.[67]	(‘H2’,): 0.0049 (‘H3’,): 0.2567 (‘H4’,): 0.6216 (‘H5’,): 0.1168	0.7126	(‘H3’,): 0.1052 (‘H4’,): 0.6977 (‘H5’,): 0.1972	0.7730	(‘H4’,): 0.6767 (‘H5’,): 0.3233	0.8308	(‘H2’,): 0.1006 (‘H3’,): 0.4022 (‘H4’,): 0.4520 (‘H5’,): 0.0452	0.6105
Dong et al.[68]	(‘H4’,): 0.0001 (‘H5’,): 0.9999	1.0000	(‘H4’,): 1.0000	0.7500	(‘H4’,): 1.0000	0.7500	(‘H2’,): 0.0039 (‘H3’,): 0.9961	0.4990
Tang et al.[69]	(‘H1’,): 0.2000 (‘H2’,): 0.2000 (‘H3’,): 0.2000 (‘H4’,): 0.2000 (‘H5’,): 0.2000	0.5000	(‘H1’,): 0.0469 (‘H2’,): 0.0469 (‘H3’,): 0.1342 (‘H4’,): 0.5615 (‘H5’,): 0.2106	0.7105	(‘H1’,): 0.0279 (‘H2’,): 0.0279 (‘H3’,): 0.0279 (‘H4’,): 0.7101 (‘H5’,): 0.2062	0.7597	(‘H1’,): 0.2000 (‘H2’,): 0.2000 (‘H3’,): 0.2000 (‘H4’,): 0.2000 (‘H5’,): 0.2000	0.5000
Li et al.[70]	(‘H3’,): 0.0035 (‘H4’,): 0.9962 (‘H5’,): 0.0003	0.7492	(‘H3’,): 0.0601 (‘H4’,): 0.7083 (‘H5’,): 0.2316	0.7929	(‘H4’,): 0.9528 (‘H5’,): 0.0472	0.7618	(‘H2’,): 0.0026 (‘H3’,): 0.6664 (‘H4’,): 0.3309 (‘H5’,): 0.0001	0.5821
This paper	(‘H3’,): 0.0223 (‘H4’,): 0.9756 (‘H5’,): 0.0021	0.7449	(‘H3’,): 0.1599 (‘H4’,): 0.5398 (‘H5’,): 0.3003	0.7851	(‘H4’,): 0.9838 (‘H5’,): 0.0162	0.7541	(‘H2’,): 0.0065 (‘H3’,): 0.4145 (‘H4’,): 0.5782 (‘H5’,): 0.0008	0.6434
Ground Truth	(‘H3’,): 0.04 (‘H4’,): 0.96	0.74	(‘H4’,): 0.84 (‘H5’,): 0.16	0.79	(‘H4’,): 1.0	0.75000	(‘H3’,): 0.44 (‘H4’,): 0.56	0.64



(a) Comparison of Final BBA Values per SubAttribute(Classical Methods)

(b) Comparison of Final BBA Values per SubAttribute(Latest Methods)

Fig. 11. Comparison of Fusion Results Across Methods.



(a) Absolute Error per SubAttribute(Classical Methods) (b) Absolute Error per SubAttribute(Latest Methods)

Fig. 12. Comparison of Absolute Errors Across Methods.

Table 7
Error statistics of different methods.

Method	Maturity error	Usability error	Recoverability error	Fault tolerance error	MAE	MaxError
Dempster	0.74000	0.04000	**0.00000**	0.64000	0.35500	0.74000
Murphy	0.02375	0.00500	0.09140	0.02750	0.03691	0.09140
Deng et al.	0.02375	0.00500	0.09110	0.03787	0.03943	0.09110
Yuan et al.	0.00994	**0.00117**	0.10094	0.05781	0.04247	0.10094
Xiao et al.	0.00921	0.00288	0.01530	0.03898	0.01659	0.03898
Song et al.	0.00926	0.00500	0.10759	0.08229	0.05104	0.10759
Dai et al.	0.00827	0.00766	0.00431	0.03735	0.00815	0.03735
Xue et al.	0.02740	0.01700	0.08083	0.02952	0.03869	0.08083
Dong et al.	0.25997	0.04000	0.00000	0.14098	0.11024	0.25997
Tang et al.	0.24000	0.07947	0.00966	0.14000	0.11728	0.24000
Li et al.	0.00827	0.00766	0.00431	0.03735	0.00815	0.03735
This paper	**0.00494**	0.00489	0.00405	**0.00335**	**0.0043**	**0.00494**

under high-conflict conditions, while the Murphy, Deng, and Yuan methods exhibit systematic underestimation.

Fig. 11 (b) further compares the proposed method with the latest evidence fusion approaches introduced in recent years in the context of software trustworthiness assessment. The Dai method produces results close to the ground truth in Maturity and Recoverability, but tends to overestimate in Fault Tolerance and underestimate in Usability. The Dong method shows considerable fluctuations, with extreme convergence in three sub-attributes and a significant underestimation in Fault Tolerance, reflecting overconfidence. Other methods consistently yield curves below the ground truth. Overall, the proposed method achieves results closest to the ground truth across all four sub-attributes.

Fig. 12 (a) and (b) present the absolute errors of classical methods versus the proposed method, as well as recent methods versus the proposed method, across the four sub-attributes. In terms of error magnitude and volatility, the proposed method yields the lowest mean absolute error (MAE) and maximum error (MaxError), while also exhibiting the smoothest error curve and the smallest variance, further validating its superior robustness and estimation accuracy under highly conflicting and uncertain scenarios.

Table 7 presents the statistical results of absolute error, mean absolute error (MAE), and maximum absolute error (MaxError) for each fusion method across four sub-attributes: maturity, usability, recoverability, and fault tolerance. The Dempster method fails in high-conflict scenarios, showing extremely high errors in maturity and fault tolerance, resulting in an MAE of 0.355 and a MaxError of 0.74. Other classical methods, such as those of Murphy, Deng, Yuan, and Song, show improvements on certain attributes but still suffer from systematic biases overall. Among the more recent approaches, Dai and Li achieve good performance on specific attributes but display fluctuations in other metrics, whereas Dong and Tang produce extreme errors and lack overall stability. In contrast, the proposed method maintains consistently low errors across all four sub-attributes. Although it does not achieve the absolute minimum in every single case, it delivers the most balanced overall performance, with an MAE of only 0.00431 and a MaxError of merely 0.00494, and without any noticeable outliers that are significantly over- or underestimated. These results further demonstrate the proposed method's superior stability and high precision in multi-source conflict evidence fusion tasks. Particularly in scenarios involving data inconsistency or noise, the method effectively mitigates extreme deviations and ensures reliable trustworthiness assessments.

This experiment confirms that the proposed method can effectively identify and adjust low-quality evidence under multi-source heterogeneous conditions, especially when dealing with conflicting, uncertain, or misleading data. Compared to state-of-the-art approaches, it delivers the highest accuracy and stability, making it well suited for complex software trustworthiness assessment scenarios.

6. Conclusion

This paper proposes a novel trustworthiness assessment model that accounts for the heterogeneity of evidence sources across different stages of the software lifecycle. Unlike traditional models that treat all evidence as a unified and homogeneous input, the proposed model explicitly classifies evidence into direct and indirect types, reflecting their differences in source and type. The model follows a hierarchical fusion structure, aggregating information progressively from the evidence layer through sub-attributes and attributes, to produce a comprehensive and interpretable trustworthiness score. To handle the conflicts that may arise during the fusion of such heterogeneous evidence, a multi-stage evidence fusion method is developed. This method measures the conflict between different types of evidence (i.e., direct and indirect) using a falsity-based approach, and further evaluates the internal consistency among evidence of the same type using credibility and uncertainty metrics. To integrate these metrics effectively, the method adopts a game-theoretic weighted combination strategy that

fuses the credibility, falsity, and uncertainty vectors into a composite discounting factor. This factor ensures a balanced correction of evidence values, enhancing both the stability and accuracy of the final trustworthiness result. Experimental validation using real software systems demonstrates that the proposed method outperforms mainstream fusion algorithms in terms of accuracy, robustness, and consistency with intuitive expectations.

Despite its demonstrated effectiveness, the proposed method has certain limitations. First, some evidence may exhibit mixed characteristics or lack sufficient metadata to support accurate classification. This ambiguity can lead to suboptimal conflict handling and may reduce the reliability of the assessment results. Second, the current method is designed for static assessment scenarios where all evidence is pre-collected and processed in batch. However, for intelligent software systems, especially those incorporating AI components, runtime parameters are constantly changing through learning, and the system state evolves accordingly. In such environments, it is difficult to collect and process real-time evidence in a timely manner. As a result, the trustworthiness evaluation may fail to reflect the actual system state, and may even be negatively influenced by outdated or unstable evidence. Future research will focus on extending the application of this method to the trustworthiness assessment of intelligent systems, addressing emerging challenges in safety, transparency, and verifiability.

CRediT authorship contribution statement

Weina Zhang: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Conceptualization, Formal analysis, Software. **Dalin Zhang:** Supervision, Methodology, Conceptualization, Project administration, Writing – original draft. **Jintao Liu:** Writing – review & editing, Resources, Funding acquisition. **Lianying He:** Investigation, Data curation, Formal analysis, Resources, Writing – review & editing.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled.

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Appendix

This appendix presents the utility theory, the conflict measurement methods, and the negation operator adopted in this paper. The utility theory based normalization process is adapted from Ref. [50], while the conflict measurement method was originally proposed in Ref. [55], and the construction of the negation operator follows the approach described in Ref. [56].

A.1. Utility theory

Let the collected metric value be denoted as F_n, and its corresponding utility value as perceived by the assessment expert be represented by $\mu(F_n)$.

Definition A.1 (*Utility Calculation for Qualitative Indicators*)**.** For a qualitative indicator α the utility value is calculated using a weighted utility function:

$$u(\alpha) = \sum_n \lambda_n \mu(F_n) \tag{A1}$$

where λ_n represents the expert's confidence in assessment level F_n, and $\mu(F_n)$ is the corresponding utility value.

Definition A.2 (*Utility Calculation for Quantitative Indicators*)**.** For a quantitative indicator β, once the piecewise utility function for β has been established, the utility value of the original assessment data $u(v_\beta)$ can be calculated using the following formula:

$$u(v_\beta) = u(F_{n+1}) - \frac{v(F_{n+1}) - v_\beta}{v(F_{n+1}) - v(F_n)} \left(u(F_{n+1}) - u(F_n)\right) \tag{A2}$$

Based on the two formulas above, the utility value of the original indicator (denoted as e) can be calculated. Then, using the following formula:

$$\beta_n = \frac{F_{n+1} - \mu_e}{F_{n+1} - F_n}, \quad \beta_{n+1} = 1 - \beta_n, \quad F_n \le \mu_e \le F_{n+1} \tag{A3}$$

The confidence levels of trustworthiness metrics with respect to the assessment utility grades F_n are established, where $u(F) = \{u(F_n), n = 0, 1, 2, \ldots, 5\}$ is provided by domain experts.

Definition A.3 (*Inverse Mapping from Mass Function to Utility*)**.** The mass function obtained by fusing the standardized evidence indicators can be converted back to the original measurement value using Eq. (A4).

$$u(m) = \sum_i \left[m_i u(H_i)\right] + m_\Omega u(H_\Omega) = \sum_i \left[m_i u(H_i)\right] + \sum_{m_i \neq 0} \left(\frac{m_\Omega u(H_i)}{|m|}\right) \tag{A4}$$

A.2. Evidence conflict measurement methods

Definition A.4 (*Shafer's Conflict Measure [53]*)**.** Shafer's conflict measure $K_{1\ldots J}$ is defined for $J > 2$ pieces of evidence $m_1, m_2 \ldots m_J$ over the same frame of discernment θ as follows:

$$K_{1\ldots J} = \sum_{\bigcap_{j=1}^{J} X_j = \emptyset,\ X_j \subseteq U} \prod_{j=1}^{J} m_j(X_j), \quad K_{1\ldots J} \in [0, 1] \tag{A5}$$

where X_j represents the hypothesis of the BBA m_j.

Definition A.5 (*Jousselme Distance [54]*)**.** Let m_1 and m_2 be two BBAs defined over a frame of discernment θ consisting of N elements. The Jousselme distance between m_1 and m_2 is defined as follows:

$$d_{\mathrm{BBA}}(m_1, m_2) = \sqrt{\frac{1}{2}(m_1 - m_2)^T D (m_1 - m_2)}, \quad d_{\mathrm{BBA}}(m_1, m_2) \in [0, 1] \tag{A6}$$

Here, m_1 and m_2 are BBAs represented in a vector form of size $2^N \times 1$. The matrix D is of size $2^N \times 2^N$, where each element is defined as $D(i, j) = \frac{|A_i \cap A_j|}{|A_i \cup A_j|}, \forall A_i, A_j \subseteq \theta$, with A_i and A_j ($for \quad i, j = 1, \ldots, 2^N$) representing the focal elements (subsets) of m_1 and m_2, respectively.

Definition A.6 (*Binary Shafer's Conflict Measurement [55]*)**.** By combining Eqs. (A5) and (A6), the total conflict $C_{1\ldots J}$ among J pieces of evidence ($J > 2$) corresponding to the BBAs $m_1, m_2, \ldots, m_J$ on the frame of discernment θ is defined as:

$$K_{1\ldots J} = \sum_{X_j \subseteq \Theta} \left\{ \left(1 - \frac{\left|\bigcap_{j=1}^{J} X_j\right|}{\left|\bigcup_{j=1}^{J} X_j\right|}\right) \prod_{j=1}^{J} m_j(X_j) \right\} \tag{A7}$$

$$C_{1\ldots J} = \frac{K_{1\ldots J} + d^{\max}_{\mathrm{BBA}_{1\ldots J}}}{2} \cdot \exp\left(-\left|\frac{K_{1\ldots J} + d^{\max}_{\mathrm{BBA}_{1\ldots J}}}{2} - 1\right|\right) \tag{A8}$$

$$d^{\max}_{\mathrm{BBA}_{1\ldots J}} = \max_{1 \le i \ne j \le J} \left\{ d_{\mathrm{BBA}}(m_i, m_j) \right\} \tag{A9}$$

Here, X_j represents any focal element of the BBA m_j. The term $\frac{\left|\bigcap_{j=1}^{J} X_j\right|}{\left|\bigcup_{j=1}^{J} X_j\right|}$ is the Jaccard similarity coefficient among all focal elements $X_j (j = 1, 2, \ldots, J)$. The term $d^{\max}_{\mathrm{BBA}_{1\ldots J}}$ denotes the maximum conflict distance between any two pieces of evidence $m_1, m_2, \ldots, m_J$, and it is introduced to mitigate the inflation of the conflict value $K_{1\ldots J}$.

A.3. Negation of BBA

Definition A.7 (*Vector Form of BBA*)**.** The vector form of the BBA (BBAV) can be written as:

$$M = (m_1, m_2, m_3, \ldots, m_{2^N})^T \tag{A10}$$

M is a column vector in a 2^N-dimensional space. Accordingly, the negated BBA vector is denoted as:

$$\overline{M} = (\overline{m}_1, \overline{m}_2, \overline{m}_3, \ldots, \overline{m}_{2^N})^T \tag{A11}$$

Definition A.8 (*Negation Operator for BBAV [56]*)**.** Given the BBAV vector M, its negation operation can be formalized as:

$$\overline{M} = G \cdot M \tag{A12}$$

where G is a negation operator matrix defined in a $2^N \times 2^N$ dimensional space.

Definition A.9 (*Construction of the Negation Operator*)**.** The negation operator $G = [g_{ij}]_{2^N \times 2^N}$ is constructed as follows: when $j \neq 2^N$, the element $g_{i,j}$ is defined as:

$$g_{i,j} = \begin{cases} 0, & i = j \\ \dfrac{\left|A_i \cap \overline{A}_j\right|}{\sum_{A_k \neq A_j,\ A_k \in 2^\theta} \left|A_k \cap \overline{A}_j\right|}, & i \neq j \end{cases} \tag{A13}$$

Where $\overline{A}_j = \theta - A_j$. When $j = 2^N, \overline{A}_j = \emptyset$, then:

$$g_{i,j} = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases} \tag{A14}$$

Data availability

Data will be made available on request.

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